

claude-windows / 6e9d0958-a2ec-4a8e-bb2d-6cce9833c527

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\subagents\agent-ad7ff344d4be89869.jsonl`  
- SHA-256: `91d634698231f2a7b81e0101038d5ed06a7d3b2d120714d48f4a0ff821c79fa3`  
- Source modified: `2026-07-03T16:41:41+00:00`  
- Imported at: `2026-07-05T16:48:26+00:00`  
- project: `subagents`  
- session_id: `6e9d0958-a2ec-4a8e-bb2d-6cce9833c527`
```

Transcript

1. user (2026-07-03T16:39:41.503Z)

You are a code-review finder agent. Review target: PR #451 of the badminton-highlight-indexer repo, checked out at C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer on branch codex/alpha-readiness-plan (HEAD cd07339). Get the diff with: git diff origin/master..HEAD (run in that directory). It is a docs-only PR: new docs/ALPHA_LAUNCH_READINESS.md (an alpha-launch readiness tracker) plus one-line index additions to docs/README.md, docs/DOC_STATUS.md, docs/NEXT_STEPS.md.

ANGLE A ? line-by-line scan. Read every added line of the diff carefully. For a docs PR, "bugs" are: internal contradictions (a claim in one section contradicted in another), wrong cross-references (a \$N reference pointing at the wrong section, a dependency table row referencing a nonexistent ID), inconsistent statuses/dates, broken markdown (tables, links), statements marked [verified] that the doc itself elsewhere undermines, and dependency-graph errors in the \$7 tracker table (e.g., A13 depends on rows that can't complete, circular deps, a row marked GATED vs TODO inconsistently with its "Gated?" column). Also check the markdown links added to docs/README.md resolve.

Return up to 6 candidate findings as a JSON array of {file, line, summary, failure_scenario}. line = line number in the new file version. failure_scenario = the concrete cost (what a reader would get wrong or what would break). Pass through every candidate with a nameable failure scenario ? do not silently drop half-believed ones. Return [] if none. Your final message must be only the JSON array.

2. assistant (2026-07-03T16:39:43.403Z)

I'll analyze PR #451 by examining the diff and checking for issues in the new documentation files. Let me start by getting the diff and reviewing the content.

3. user (2026-07-03T16:39:44.274Z)

```
diff --git a/docs/ALPHA_LAUNCH_READINESS.md b/docs/ALPHA_LAUNCH_READINESS.md  
new file mode 100644  
index 0000000..4b3d49d  
--- /dev/null  
+++ b/docs/ALPHA_LAUNCH_READINESS.md  
@@ -0,0 +1,171 @@  
+# Alpha Launch Readiness Plan  
+  
+> **Status:** `DRAFT` - cross-repo execution tracker; **Owner:** @avidullu; **Created:** 2026-07-03; **Last updated:** 2026-07-03  
+> **Lifecycle:** `DRAFT -> IN PROGRESS -> DONE -> archived`  
+> **Tracking anchors:** Section 7 is the progress tracker and source of truth; indexed in `docs/README.md` and `docs/DOC_STATUS.md`.  
+> **Relation to existing docs:** consolidates the launch-relevant pending items from `NEXT_STEPS.md`, `ROADMAP.md`, `data_pipeline/GOLDEN_VIDEOS.md`, `R2_EVAL_METRIC_DESIGN.md`, `COMPUTE_DECOUPLED_SERVING/`, and the sibling `khelsutra` Studio docs.
```

```

+> **Honesty note:** `[verified]` means checked against the local synced repos on 2026-07-03.
`[analysis]` means derived from a local command run on that snapshot. This is not legal,
financial, or launch approval.
+
+---
+
+### 0. TL;DR
+
+The corpus is now large enough to move from "grow labels first" into alpha-readiness work, but
the 15-video headline is not itself a launch gate pass. `[verified]` The golden manifest has 15
badminton videos; `[analysis]` the local manifest needed path rebasing before manifest-driven
commands ran; `[analysis]` only 6 of 15 feature JSONs had the full fusion schema; `[analysis]`
the n=15 quality baseline is stale and must be intentionally refreshed before it can gate
anything.
+
+The first controlled alpha should be a hosted, authenticated product loop: upload a video,
process it, select rallies, compile, and download a reel. The owned-model/commercial rally-
detection claim remains a separate NO-GO until the ship-gate target is ratified and the WASB
checkpoint provenance is resolved.
+
+### 1. Problem & goal
+
+The previous launch blockers were spread across roadmap docs and mixed together three different
questions:
+
+- **Can an invited tester use the product without SSH?**
+- **Can the 15-video golden corpus now support honest quality gates?**
+- **Can we claim or ship an owned/commercial rally detector?**
+
+This doc separates those into executable workstreams. The goal is to reach an invite-only alpha
where the product is usable end-to-end, the quality claims are honest and reproducible, and the
remaining model/legal gates are explicit rather than accidentally implied by launch activity.
+
+### 2. Decisions locked
+
+| # | Decision | Source / date | Implication |
+|---|---|---|---|
+| D1 | **Product alpha and owned-model launch are separate decisions.** | `NEXT_STEPS.md` NO-GO
banner, 2026-06-30; verified 2026-07-03 | We can launch a controlled hosted product loop without
claiming the owned model is commercially cleared. |
+| D2 | **First alpha must be page-driven, not SSH/CLI-driven.** | `NEXT_STEPS.md` P0 UI-
driven/shareable item; verified 2026-07-03 | Hosted service, edge auth, upload, job polling,
compile, and download are launch-critical. |
+| D3 | **The 15-video corpus is now a measurement asset, not a free launch pass.** |
`data_pipeline/GOLDEN_VIDEOS.md`; local eval runs, 2026-07-03 | Rebase/portability, feature
completeness, baseline refresh, and gate ratification come before quality claims. |
+| D4 | **Do not promote served Gate A by default from the current n=15 evidence.** |
`backend.eval.serve_contrast` run, 2026-07-03 | Proposal-side lift does not survive served-
windowing safety; keep it as an opt-in/analysis lever. |
+| D5 | **P10/P11 corpus promotion and train/re-eval are alpha-plus unless owner reopens D13.**
| sibling `khelsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`; verified 2026-07-03 | First alpha can use
manual/offline corpus promotion; automated flywheel follows after serving and gates are stable.
|
+
+### 3. Verified snapshot, 2026-07-03
+
+All five Git checkouts were fast-forwarded before analysis:
+
+| Repo | Branch | Synced HEAD |
+|---|---|---|
+| `badminton-highlight-indexer` | `master` | `e4914d1` |
+| `khelsutra` | `master` | `b0f70a5` |
+| `rally-annotator` | `main` | `ef9e088` |
+| `sports-data-collector` | `master` | `bfc014f` |
+| `sports-obsreport` | `main` | `a0c7252` |

```

```

+
+Corpus/eval facts:
+
+- `[verified]` `docs/data_pipeline/GOLDEN_VIDEOS.md` says `output/human_lovo_manifest.json` is
the ingested golden corpus at `n = 15`.
+- `[analysis]` The local manifest had stale absolute paths. A temporary rebased manifest found
all 15 trajectories and labels in this workspace.
+- `[analysis]` There are 15 `output/*_golden_features.json` files, but only 6 carry full
shuttle/person/audio candidate features. The newer 9 are windowing-only schema and are skipped
by fusion/person eval.
+- `[analysis]` `python -m backend.eval.fusion_compare --features-dir output` loaded only 6
videos and measured person-fusion `Delta F1 = -0.021`; not promotable.
+- `[analysis]` `python -m backend.eval.serve_contrast --manifest <rebased-n15>` measured
proposal Gate A mean `Delta F1 +0.0287`, but served mean `Delta F1 -0.0328`; not served-safe.
+- `[analysis]` `python -m training.gen0.harness --manifest <rebased-n15> --floor
eval_baselines/heuristic_lovo_n6.json` produced learned LOVO mean F1 `0.551`, `ship=False`;
blockers remain stale n=6 floor/sign test, uncalibrated ship target, and WASB C3 provenance.
+- `[analysis]` `python scripts/nightly_regression.py --manifest <rebased-n15> --features-dir
output --no-report` reported default/milestone tolF1 `0.3636` vs baseline-off tolF1 `0.1382`,
but marked the frozen baseline stale due to config/corpus changes.
+- `[analysis]` `python -m backend.eval.ablation --features-dir output --granularity group`
currently fails at import due to a circular import between `backend/eval/ablation.py` and
`backend/eval/ablation_pdf.py`.
+
+## 4. Launch strategy
+
+### 4.1 Alpha lane: make the product usable
+
+The alpha lane is successful when an invited tester can open the app, authenticate, upload a
video, watch the job progress, pick rallies, compile a reel, and download it. This lane should
avoid model-quality promises beyond the measured state and should default to the most reliable
serving path.
+
+### 4.2 Quality lane: make the 15-video corpus gate-ready
+
+The quality lane is successful when the 15-video corpus is portable, reproducible, feature-
complete, and backed by refreshed baselines. Only then should R2 tolF1, Gen-0/TemporalMaxer, or
other model/promotion decisions be ratified.
+
+### 4.3 Flywheel lane: make tester feedback useful
+
+The flywheel lane is successful when user corrections can flow into the collector/vault path
with consent/provenance and a visible re-eval delta. This is valuable for alpha learning, but it
should not block the first controlled alpha if manual promotion can bridge the gap.
+
+## 5. Risks and guardrails
+
+| Risk | Why it matters | Guardrail |
+|---|---|---|
+| Launch activity implies model launch approval | The engine docs explicitly say rally-
detection product launch is NO-GO until ship target and WASB C3 clear | Keep all alpha copy
scoped to invite-only processing, not owned-model claims |
+| Stale local manifests create false confidence | Manifest-driven tools skipped or failed until
paths were rebased | Make the corpus manifest portable and add a path/fixture smoke check |
+| 15-video corpus is only partially feature-complete | Fusion/person/audio eval is still n=6
today | Regenerate or upconvert full-fusion features for all 15 before fusion decisions |
+| Baseline refresh accidentally rubber-stamps a regression | Nightly runner correctly reported
stale baselines | Refresh baselines only after reviewer sign-off and record the command/output |
+| P10/P11 expands scope before hosted alpha is usable | Corpus flywheel is valuable but can
consume the alpha launch window | Keep P10/P11 behind explicit owner reopen unless alpha lane is
already stable |
+| WASB checkpoint provenance remains unresolved | Blocks commercial sharing of a trained model
regardless of F1 | Treat C3 as a hard model-release gate; do not hide it inside product launch
tasks |
+

```

6. Honest limits

+
+Completing this doc's alpha lane does ****not**** prove the owned model is ready, that WASB weights are commercially clean, that the product can serve unbounded public traffic, that multi-sport is ready, or that every tester correction automatically improves the model.
+
+Completing the quality lane does ****not**** by itself deploy a service. It only makes the measurement surface honest enough for a launch decision.

7. Deliverables & progress tracker

+
+Legend: TODO = not started, IN PROGRESS = actively being worked, DONE = shipped/verified, GATED = blocked on owner/legal/platform decision. One PR per row unless the row explicitly names a cross-repo batch.

+
+| ID | Deliverable | Repo / owner | Depends on | Gated? | Status | PR |
+|----|-----|-----|-----|-----|-----|----|
+| A0 | Create this alpha readiness tracker and index it in docs | `badminton-highlight-indexer` | - | No | IN PROGRESS | #451 |
+| A1 | Stand up alpha serving endpoint: Cloud Run/GPU or approved host, health check, edge auth ON, upload -> process -> jobs -> compile -> download verified | `khesutra` + `badminton-highlight-indexer` | A0 | Owner spend / CF config | TODO | - |
+| A2 | Make the n=15 golden manifest portable/reproducible; add a path/corpus smoke check that fails loudly when labels or trajectories are missing | `badminton-highlight-indexer` | A0 | No | TODO | - |
+| A3 | Promote the 15-video source-video/MD5 records into the collector/vault path; reconcile collector's six-video source manifest with the 15-video eval corpus | `sports-data-collector` + vault | A2 | Owner upload/storage scope | TODO | - |
+| A4 | Refresh the n=15 nightly regression baseline intentionally and record the exact command/output; keep stale-baseline warnings until reviewed | `badminton-highlight-indexer` | A2 | Reviewer sign-off | TODO | - |
+| A5 | Recompute the heuristic n=15 floor and replace the stale `heuristic\_lovo\_n6` floor for future Gen-0 comparisons | `badminton-highlight-indexer` | A2 | No | TODO | - |
+| A6 | Fix `backend.eval.ablation` CLI circular import, then run the R2/tolF1 ablation on the n=15 corpus | `badminton-highlight-indexer` | A2, A4 | No | TODO | - |
+| A7 | Regenerate or upconvert full-fusion feature JSONs for the newer 9 videos so fusion/person/audio analyses run over all 15 | `badminton-highlight-indexer` | A2 | GPU/data availability | TODO | - |
+| A8 | Re-run `fusion\_compare`, group ablation, and served contrast after A7; explicitly promote/reject person fusion, Gate A served-default, and any other R-series default flips | `badminton-highlight-indexer` | A6, A7 | Reviewer sign-off | TODO | - |
+| A9 | Run Gen-0/TemporalMaxer shadow training against the refreshed n=15 floor; produce a non-serving artifact bundle and gate verdict | `badminton-highlight-indexer` | A5, A8 | Owner approval for M0.4 scope | TODO | - |
+| A10 | Decide and record the alpha ship-gate target: metric, corpus slice, minimum threshold, significance rule, and what claims are allowed | Owner + `badminton-highlight-indexer` | A4, A5, A8 | Owner decision | GATED | - |
+| A11 | Resolve or explicitly defer WASB C3 provenance for alpha messaging; if deferred, document "no commercial model sharing" copy guardrail | Owner/legal | A10 | Legal/provenance | GATED | - |
+| A12 | Reopen or defer P10/P11 corpus flywheel: `PATCH /api/segments`, collector promotion, `train|reeval` job, and delta-F1 surface | `khesutra` + `badminton-highlight-indexer` + collector | A1, A3 | Owner D13 reopen | GATED | - |
+| A13 | Alpha launch review: runbook, privacy/consent copy, known limits, rollback/disable steps, and owner GO/NO-GO | Cross-repo | A1, A4, A10, A11 | Owner launch decision | TODO | - |

8. Concrete next PR sequence

- +1. ****Docs PR:**** land this tracker.
- +2. ****Corpus portability PR:**** rebase/portable manifest tooling plus missing-file smoke test.
- +3. ****Hosted alpha PR/batch:**** health/auth/upload/process/job/compile/download verification and runbook updates.
- +4. ****Baseline PR:**** intentional n=15 nightly baseline refresh and heuristic floor refresh.
- +5. ****Ablation/fusion PRs:**** fix ablation import, complete 15-video features, rerun promotion/rejection decisions.

```

+6. **Launch decision PR/docs update:** record ship-gate target, C3 status, alpha copy limits,
and launch checklist.
+
+### 9. Open questions
+
+1. **Owner:** Is the first alpha allowed to use manual/offline corpus promotion, or should
P10/P11 be reopened before any tester wave?
+2. **Owner/platform:** Which hosted compute target is the next stand-up: Cloud Run GPU, GCE L4
VM, or another approved GPU host?
+3. **Owner/legal:** What is the acceptable alpha wording while WASB C3 remains unresolved?
+4. **Engineering:** Should the portable corpus manifest live as a generated artifact, a
checked-in redacted manifest, or both?
+5. **Engineering:** Should n=15 feature regeneration target full fusion schema for every clip,
or should the eval loaders support mixed schemas with explicit capability tags?
+
+### 10. Definition of done
+
+- [ ] Invited tester can complete the hosted record/upload -> reel download CUJ without SSH.
+- [ ] Alpha service is gated behind the chosen auth posture and has a documented
disable/rollback path.
+- [ ] n=15 corpus is portable enough that manifest-driven quality commands do not rely on stale
local absolute paths.
+- [ ] n=15 nightly baseline and heuristic floor are refreshed with recorded commands.
+- [ ] R2/toIF1 gate is promoted or rejected with a recorded ablation result.
+- [ ] Fusion/person/Gate-A served-default decisions are explicitly promoted or rejected on the
15-video corpus.
+- [ ] Ship-gate target and allowed alpha claims are recorded.
+- [ ] WASB C3 status is recorded and reflected in alpha messaging.
+- [ ] P10/P11 are either reopened with a committed sequence or explicitly deferred for alpha-
plus.
+- [ ] Owner records GO/NO-GO for the first alpha wave.
+
+### 11. References
+
+Internal docs:
+
+- `docs/NEXT_STEPS.md`
+- `docs/ROADMAP.md`
+- `docs/data_pipeline/GOLDEN_VIDEOS.md`
+- `docs/R2_EVAL_METRIC_DESIGN.md`
+- `docs/RALLY_DETECTION_FIXES_PLAN.md`
+- `docs/COMPUTE_DECOUPLED_SERVING/README.md`
+- sibling `khehsutra/NEXT_STEPS.md`
+- sibling `khehsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`
+- sibling `khehsutra/docs/PER_RALLY_SELECTION.md`
+- sibling `sports-data-collector/data/golden_source_videos.json`
+
+Local verification commands used for the 2026-07-03 snapshot:
+
+```powershell
+python -m backend.eval.fusion_compare --features-dir output
+python -m backend.eval.serve_contrast --manifest <rebased-n15-manifest>
+python -m training.gen0.harness --manifest <rebased-n15-manifest> --floor
eval_baselines\heuristic_lovo_n6.json --version analysis-n15 --out <temp-out>
+python scripts\nightly_regression.py --manifest <rebased-n15-manifest> --features-dir output
--no-report
+python -m backend.eval.ablation --features-dir output --granularity group
+```
+
+---
+
+### Changelog
+
+- 2026-07-03 - Created from a synced five-repo review of `NEXT_STEPS`, `ROADMAP`, golden-corpus
docs, local eval artifacts, and the current 15-video corpus state.

```

```

diff --git a/docs/DOC_STATUS.md b/docs/DOC_STATUS.md
index 34411fd..2541672 100644
--- a/docs/DOC_STATUS.md
+++ b/docs/DOC_STATUS.md
@@ -30,6 +30,7 @@ Before a foundational/tracked doc moves to `docs/archives/`:
### 3a. Tracked projects ? lifecycle docs (archive when DONE; each doc's $7 is the completeness
source-of-truth)
| Doc | Status | Archive-ready? | Next action |
| ---| ---| ---| ---|
+| `ALPHA_LAUNCH_READINESS.md` | **DRAFT** ? cross-repo alpha readiness tracker created
2026-07-03 after the golden corpus reached n=15; separates hosted product-alpha work from owned-
model/commercial rally-detection gates | No | Execute $7 in order: corpus portability + n=15
baselines, hosted alpha serving, ablation/fusion reruns, then launch GO/NO-GO |
| `COMPUTE_DECOUPLED_SERVING/` | **IN PROGRESS** ? M0 approved + **M1?M9 LARGELY SHIPPED**
(#279?#290; M5 #286, M6 #287, M8 #288, M9-auth #290 all merged, default-OFF); Gen-0 weights
pushed (`weights/0.1.0-l4/`) | No | M11 flywheel + M12 gdrive-api (owner residual:
counsel/prices/OAuth/real-run within the D17 cap) |
| `RALLY_QUALITY_RESEARCH.md` | Foundation, **tracked & ACTIVE** ($7) | No | Drives the
R-series; complete when $7.1 single quality number is attributed |
| `R2_EVAL_METRIC_DESIGN.md` | IMPLEMENTED (augmenting) | No (gates R4 + dual-eval; ablation
pending) | Promote-or-reject tolF1 as the gate (ablation) ? then archive with the R-cluster |
diff --git a/docs/NEXT_STEPS.md b/docs/NEXT_STEPS.md
index f186ff9..a9f35a8 100644
--- a/docs/NEXT_STEPS.md
+++ b/docs/NEXT_STEPS.md
@@ -9,7 +9,7 @@
## ? Prioritized pending ? all tracks
**P0 ? highest leverage / unblocks the most**
-0. **? [owner-priority, 2026-06-21] MAKE IT UI-DRIVEN + ALPHA-SHAREABLE ? "a few recordings
lying around ? highlights, from a page, no SSH". ** Today the cloud flow is **SSH/CLI-only**
(`docs/CLOUD_GPU_DRYRUN_GUIDE.md`); **M12 does NOT fix it. ** Needs (a separate khelsutra `web/`
SPA + deploy track, NOT an engine seam): **i) deploy the engine as a reachable **service**
(M7 ? Cloud Run service, ? owner GCP spend for a *persistent* endpoint), **ii) the **SPA
ingest/progress screen** (khelsutra `web/` B-P2: enter/upload video ? `/api/process` ? poll
`/api/jobs` ? reel; ZERO engine code), **iii) flip **M9 edge-auth ON** (merged #290, default-
OFF) to gate alpha tenants, **iv) a **compute PICKER in the UI** ("your machine vs our
cloud", D13/D15). The engine seams make it *possible*; the SPA makes it *usable*. Full record:
[[ui-driven-cloud-serving-gap]].
+0. **? [owner-priority, 2026-06-21] MAKE IT UI-DRIVEN + ALPHA-SHAREABLE ? "a few recordings
lying around ? highlights, from a page, no SSH". ** Today the cloud flow is **SSH/CLI-only**
(`docs/CLOUD_GPU_DRYRUN_GUIDE.md`); **M12 does NOT fix it. ** Needs (a separate khelsutra `web/`
SPA + deploy track, NOT an engine seam): **i) deploy the engine as a reachable **service**
(M7 ? Cloud Run service, ? owner GCP spend for a *persistent* endpoint), **ii) the **SPA
ingest/progress screen** (khelsutra `web/` B-P2: enter/upload video ? `/api/process` ? poll
`/api/jobs` ? reel; ZERO engine code), **iii) flip **M9 edge-auth ON** (merged #290, default-
OFF) to gate alpha tenants, **iv) a **compute PICKER in the UI** ("your machine vs our
cloud", D13/D15). The engine seams make it *possible*; the SPA makes it *usable*. Full record:
[[ui-driven-cloud-serving-gap]]. **Concrete execution tracker after the n=15 corpus review:**
`docs/ALPHA_LAUNCH_READINESS.md` $7.
0b. **? [packaging track, IN PROGRESS ? 11 PRs merged 2026-06-21/22] Downloadable batteries-
included app ? `docs/PACKAGING_PLAN.md`. ** The LIGHT-tier engine is bundle-ready: `pip install`
? `indexer --app` serves the **bundled SPA** single-origin, **torch-free**, with app-home state
/ ffmpeg resolver / DNS-rebinding guard / single-instance lock / safe DB upgrades (P0a?e, P1a/b,
P2-launcher/P2a/P2d/P2e). **? Next (fresh focused pushes): ** (1) **P3 GPU validation +
screencast** (L4, ~$0.50, DELETE-after, $100 cap ? `[[gcp-vm-always-delete]]`; ? native-WASB
0-rally tuned-config blocker), ** (2) **P2b-install (uv/script + truly-local default config),
** (3) **P2c first-run wizard (cross-repo khelsutra SPA from `/api/capabilities`), then P3-cloud
(Cloud Run) + P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` / `[[next-
session-packaging-app]]`.
1. ? **COMPUTE_DECOUPLED_SERVING` M0?M9 LARGELY SHIPPED** (2026-06-21, ADR-014 ratified).
M1?M4 (#279/#280/#282/#283), **M5 #286, M6 #287, M8 #288, M9-auth #290** all merged (default-
OFF); the **real M7 L4 run** validated the worker (22 rallies, VM deleted, $0). All $8 gating Qs
locked (D7?D17). **Owner residual** (not blocking Claude's default-OFF code): counsel ToS/DPA
(M9-consent live + M11 live), real GCP prices+FX (M8 calibrate), Google OAuth app (M12 real), CF

```

team-domain/AUD + ingress-lock; real cloud verification authorized within a **\$100/?10k cap** (D17).

2. **[owner, ongoing] Grow the golden corpus** ? multi-court badminton + ?3 matches/sport (TT first). The single highest-leverage move; the quality track *and* full-corpus training feed on it.

```
diff --git a/docs/README.md b/docs/README.md
index 0df1a69..58be5ff 100644
--- a/docs/README.md
+++ b/docs/README.md
@@ -15,6 +15,7 @@
- [DOC_STATUS.md](DOC_STATUS.md) ? **? living Documentation Status & Archival Register**: every foundational/tracked doc's lifecycle + archive-readiness, the **archival due-diligence checklist** (*honestly update ? then archive*), and the open doc-hygiene findings. Review here; update rows as docs complete.
```

Product / platform plans

+ ALPHA_LAUNCH_READINESS.md ? **? Active tracked project (`DRAFT`, created 2026-07-03):** cross-repo alpha readiness tracker after the golden corpus reached 15 videos. Separates the hosted product-alpha lane (shareable upload?process?pick?reel) from the owned-model/commercial rally-detection gate (ship target + WASB-C3 still unresolved). §7 is the source of truth for the concrete next PR sequence: corpus portability, n=15 baseline refresh, ablation/fusion reruns, hosted serving, alpha copy/launch review.

- [COMPUTE_DECOUPLED_SERVING/](COMPUTE_DECOUPLED_SERVING/README.md) ? **? Active tracked project (2026-06-21, `DRAFT` owner-gated):** the productionization successor to DECOUPLED_COMPUTE (its deferred M3/M4). **One pure core (DETECT?WINDOW?STITCH) that never branches on deployment profile**, with a `ComputeBackend` (where it runs) + `StorageBackend` (what bytes) selected by **one startup config**. Ships **truly-local** (local/USB + local GPU, in-process stitch, zero cloud) + **cloud-serving** (upload<2GB / gdrive / bucket ? a **Cloud Run GPU job runs detect AND stitch**, metered into a per-job cost ledger) + **cpu-mock** (CI). Includes [TESTING_STRATEGY.md](COMPUTE_DECOUPLED_SERVING/TESTING_STRATEGY.md) (profile-PARITY proof for \$0), [architecture.svg](COMPUTE_DECOUPLED_SERVING/architecture.svg), a [runlogs/](COMPUTE_DECOUPLED_SERVING/runlogs/) posterity dir, and **ADR-014** ([lineage ADR-009?010?011?012?013?014](archives/decisions/DECISIONS.md)). §7 M0?M10 tracker is the source of truth.

- [DECOUPLED_COMPUTE/](archives/past_projects/DECOUPLED_COMPUTE/README.md) ? ? **DELIVERED + ARCHIVED (2026-06-21).** **Lifted the rally pipeline onto a **rented cloud-native GPU** doing bucket-driven `train` + `infer`: storage/compute seams (PRs #246?#259) + the **M3.5 native WASB runner**; **M0?M2 + M3.5 proven on a GCP L4** (M2 ? `weights/0.1.0-l4/`). Delivered on **GCP** ? DigitalOcean dropped for compute ([ADR-013](archives/decisions/DECISIONS.md)). **M3/M4 (serving endpoint + per-video billing + scale-to-zero) deferred ? the "Cloud Run + GPU serving" subproject** ([ADR-012](archives/decisions/DECISIONS.md)). Snapshot under `docs/archives/past\_projects/DECOUPLED\_COMPUTE/` (incl. `DEPLOY\_REPORT\_GCP\_2026-06-20.md`).

- PACKAGING_PLAN.md ? **? Active tracked project (`IN PROGRESS`, 11 PRs merged 2026-06-21/22):** ship the engine as a **downloadable batteries-included Python app** ? `pip install` ? `indexer --app` boots a local webserver and opens the **bundled KhelSutra SPA** single-origin, **torch-free** (LIGHT tier), cross-platform, no native component (decision D10/01). Built ON the decoupled-compute seam. **Supersedes the `DESKTOP\_APP\_PLAN.md` narrative.** §7 tracker = source of truth (P0a?e ?, P1a/b ?, P2-launcher/P2a/P2d/P2e ?). **? Next:** P3 GPU validation + screencast ? P2b-install ? P2c wizard (cross-repo) ? P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` / `[[next-session-packaging-app]]`.

4. assistant (2026-07-03T16:39:46.247Z)

Now let me read the full content to carefully check for the issues mentioned in the angle A review criteria:

5. user (2026-07-03T16:39:46.959Z)

```
1      # Alpha Launch Readiness Plan
2
3      > **Status:** `DRAFT` - cross-repo execution tracker; **Owner:** @avidullu; **Created:** 2026-07-03; **Last updated:** 2026-07-03
4      > **Lifecycle:** `DRAFT` -> IN PROGRESS -> DONE -> archived`
5      > **Tracking anchors:** Section 7 is the progress tracker and source of truth; indexed
```

```

in `docs/README.md` and `docs/DOC_STATUS.md`.
6     > **Relation to existing docs:** consolidates the launch-relevant pending items from
`NEXT_STEPS.md`, `ROADMAP.md`, `data_pipeline/GOLDEN_VIDEOS.md`, `R2_EVAL_METRIC_DESIGN.md`,
`COMPUTE_DECOUPLED_SERVING/`, and the sibling `khelsutra` Studio docs.
7     > **Honesty note:** `[verified]` means checked against the local synced repos on
2026-07-03. `[analysis]` means derived from a local command run on that snapshot. This is not
legal, financial, or launch approval.
8
9     ---
10
11     ## 0. TL;DR
12
13     The corpus is now large enough to move from "grow labels first" into alpha-readiness
work, but the 15-video headline is not itself a launch gate pass. `[verified]` The golden
manifest has 15 badminton videos; `[analysis]` the local manifest needed path rebasing before
manifest-driven commands ran; `[analysis]` only 6 of 15 feature JSONs had the full fusion
schema; `[analysis]` the n=15 quality baseline is stale and must be intentionally refreshed
before it can gate anything.
14
15     The first controlled alpha should be a hosted, authenticated product loop: upload a
video, process it, select rallies, compile, and download a reel. The owned-model/commercial
rally-detection claim remains a separate NO-GO until the ship-gate target is ratified and the
WASB checkpoint provenance is resolved.
16
17     ## 1. Problem & goal
18
19     The previous launch blockers were spread across roadmap docs and mixed together three
different questions:
20
21     - **Can an invited tester use the product without SSH?**
22     - **Can the 15-video golden corpus now support honest quality gates?**
23     - **Can we claim or ship an owned/commercial rally detector?**
24
25     This doc separates those into executable workstreams. The goal is to reach an invite-
only alpha where the product is usable end-to-end, the quality claims are honest and
reproducible, and the remaining model/legal gates are explicit rather than accidentally implied
by launch activity.
26
27     ## 2. Decisions locked
28
29     | # | Decision | Source / date | Implication |
30     |---|---|---|---|
31     | D1 | **Product alpha and owned-model launch are separate decisions.** |
`NEXT_STEPS.md` NO-GO banner, 2026-06-30; verified 2026-07-03 | We can launch a controlled
hosted product loop without claiming the owned model is commercially cleared. |
32     | D2 | **First alpha must be page-driven, not SSH/CLI-driven.** | `NEXT_STEPS.md` P0 UI-
driven/shareable item; verified 2026-07-03 | Hosted service, edge auth, upload, job polling,
compile, and download are launch-critical. |
33     | D3 | **The 15-video corpus is now a measurement asset, not a free launch pass.** |
`data_pipeline/GOLDEN_VIDEOS.md`; local eval runs, 2026-07-03 | Rebase/portability, feature
completeness, baseline refresh, and gate ratification come before quality claims. |
34     | D4 | **Do not promote served Gate A by default from the current n=15 evidence.** |
`backend.eval.serve_contrast` run, 2026-07-03 | Proposal-side lift does not survive served-
windowing safety; keep it as an opt-in/analysis lever. |
35     | D5 | **P10/P11 corpus promotion and train/re-eval are alpha-plus unless owner reopens
D13.** | sibling `khelsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`; verified 2026-07-03 | First alpha
can use manual/offline corpus promotion; automated flywheel follows after serving and gates are
stable. |
36
37     ## 3. Verified snapshot, 2026-07-03
38
39     All five Git checkouts were fast-forwarded before analysis:
40
41     | Repo | Branch | Synced HEAD |
42     |---|---|---|

```



```

43 | `badminton-highlight-indexer` | `master` | `e4914d1` |
44 | `khelsutra` | `master` | `b0f70a5` |
45 | `rally-annotator` | `main` | `ef9e088` |
46 | `sports-data-collector` | `master` | `bfc014f` |
47 | `sports-obsreport` | `main` | `a0c7252` |
48
49 Corpus/eval facts:
50
51 - `[verified]` `docs/data_pipeline/GOLDEN_VIDEOS.md` says
`output/human_lovo_manifest.json` is the ingested golden corpus at `n = 15`.
52 - `[analysis]` The local manifest had stale absolute paths. A temporary rebased manifest
found all 15 trajectories and labels in this workspace.
53 - `[analysis]` There are 15 `output/*_golden_features.json` files, but only 6 carry full
shuttle/person/audio candidate features. The newer 9 are windowing-only schema and are skipped
by fusion/person eval.
54 - `[analysis]` `python -m backend.eval.fusion_compare --features-dir output` loaded only
6 videos and measured person-fusion `Delta F1 = -0.021`; not promotable.
55 - `[analysis]` `python -m backend.eval.serve_contrast --manifest <rebased-n15>` measured
proposal Gate A mean `Delta F1 +0.0287`, but served mean `Delta F1 -0.0328`; not served-safe.
56 - `[analysis]` `python -m training.gen0.harness --manifest <rebased-n15> --floor
eval_baselines/heuristic_lovo_n6.json` produced learned LOVO mean F1 `0.551`, `ship=False`;
blockers remain stale n=6 floor/sign test, uncalibrated ship target, and WASB C3 provenance.
57 - `[analysis]` `python scripts/nightly_regression.py --manifest <rebased-n15>
--features-dir output --no-report` reported default/milestone tolF1 `0.3636` vs baseline-off
tolF1 `0.1382`, but marked the frozen baseline stale due to config/corpus changes.
58 - `[analysis]` `python -m backend.eval.ablation --features-dir output --granularity
group` currently fails at import due to a circular import between `backend/eval/ablation.py` and
`backend/eval/ablation_pdf.py`.
59
60 ## 4. Launch strategy
61
62 ### 4.1 Alpha lane: make the product usable
63
64 The alpha lane is successful when an invited tester can open the app, authenticate,
upload a video, watch the job progress, pick rallies, compile a reel, and download it. This lane
should avoid model-quality promises beyond the measured state and should default to the most
reliable serving path.
65
66 ### 4.2 Quality lane: make the 15-video corpus gate-ready
67
68 The quality lane is successful when the 15-video corpus is portable, reproducible,
feature-complete, and backed by refreshed baselines. Only then should R2 tolF1,
Gen-0/TemporalMaxer, or other model/promotion decisions be ratified.
69
70 ### 4.3 Flywheel lane: make tester feedback useful
71
72 The flywheel lane is successful when user corrections can flow into the collector/vault
path with consent/provenance and a visible re-eval delta. This is valuable for alpha learning,
but it should not block the first controlled alpha if manual promotion can bridge the gap.
73
74 ## 5. Risks and guardrails
75
76 | Risk | Why it matters | Guardrail |
77 |---|---|---|
78 | Launch activity implies model launch approval | The engine docs explicitly say rally-
detection product launch is NO-GO until ship target and WASB C3 clear | Keep all alpha copy
scoped to invite-only processing, not owned-model claims |
79 | Stale local manifests create false confidence | Manifest-driven tools skipped or
failed until paths were rebased | Make the corpus manifest portable and add a path/fixture smoke
check |
80 | 15-video corpus is only partially feature-complete | Fusion/person/audio eval is still
n=6 today | Regenerate or upconvert full-fusion features for all 15 before fusion decisions |
81 | Baseline refresh accidentally rubber-stamps a regression | Nightly runner correctly
reported stale baselines | Refresh baselines only after reviewer sign-off and record the
command/output |

```

82 | P10/P11 expands scope before hosted alpha is usable | Corpus flywheel is valuable but
can consume the alpha launch window | Keep P10/P11 behind explicit owner reopen unless alpha
lane is already stable |

83 | WASB checkpoint provenance remains unresolved | Blocks commercial sharing of a trained
model regardless of F1 | Treat C3 as a hard model-release gate; do not hide it inside product
launch tasks |

84

85 ## 6. Honest limits

86

87 Completing this doc's alpha lane does ****not**** prove the owned model is ready, that WASB
weights are commercially clean, that the product can serve unbounded public traffic, that multi-
sport is ready, or that every tester correction automatically improves the model.

88

89 Completing the quality lane does ****not**** by itself deploy a service. It only makes the
measurement surface honest enough for a launch decision.

90

91 ## 7. Deliverables & progress tracker

92

93 Legend: TODO = not started, IN PROGRESS = actively being worked, DONE =
shipped/verified, GATED = blocked on owner/legal/platform decision. One PR per row unless the
row explicitly names a cross-repo batch.

94

95 | ID | Deliverable | Repo / owner | Depends on | Gated? | Status | PR |

96 |----|-----|-----|-----|-----|-----|----|

97 | A0 | Create this alpha readiness tracker and index it in docs | `badminton-highlight-
indexer` | - | No | IN PROGRESS | #451 |

98 | A1 | Stand up alpha serving endpoint: Cloud Run/GPU or approved host, health check,
edge auth ON, upload -> process -> jobs -> compile -> download verified | `khelsutra` +
`badminton-highlight-indexer` | A0 | Owner spend / CF config | TODO | - |

99 | A2 | Make the n=15 golden manifest portable/reproducible; add a path/corpus smoke
check that fails loudly when labels or trajectories are missing | `badminton-highlight-indexer`
| A0 | No | TODO | - |

100 | A3 | Promote the 15-video source-video/MD5 records into the collector/vault path;
reconcile collector's six-video source manifest with the 15-video eval corpus | `sports-data-
collector` + vault | A2 | Owner upload/storage scope | TODO | - |

101 | A4 | Refresh the n=15 nightly regression baseline intentionally and record the exact
command/output; keep stale-baseline warnings until reviewed | `badminton-highlight-indexer` | A2
| Reviewer sign-off | TODO | - |

102 | A5 | Recompute the heuristic n=15 floor and replace the stale `heuristic\_lovo\_n6`
floor for future Gen-0 comparisons | `badminton-highlight-indexer` | A2 | No | TODO | - |

103 | A6 | Fix `backend.eval.ablation` CLI circular import, then run the R2/toIF1 ablation
on the n=15 corpus | `badminton-highlight-indexer` | A2, A4 | No | TODO | - |

104 | A7 | Regenerate or upconvert full-fusion feature JSONs for the newer 9 videos so
fusion/person/audio analyses run over all 15 | `badminton-highlight-indexer` | A2 | GPU/data
availability | TODO | - |

105 | A8 | Re-run `fusion\_compare`, group ablation, and served contrast after A7; explicitly
promote/reject person fusion, Gate A served-default, and any other R-series default flips |
`badminton-highlight-indexer` | A6, A7 | Reviewer sign-off | TODO | - |

106 | A9 | Run Gen-0/TemporalMaxer shadow training against the refreshed n=15 floor; produce
a non-serving artifact bundle and gate verdict | `badminton-highlight-indexer` | A5, A8 | Owner
approval for M0.4 scope | TODO | - |

107 | A10 | Decide and record the alpha ship-gate target: metric, corpus slice, minimum
threshold, significance rule, and what claims are allowed | Owner + `badminton-highlight-
indexer` | A4, A5, A8 | Owner decision | GATED | - |

108 | A11 | Resolve or explicitly defer WASB C3 provenance for alpha messaging; if deferred,
document "no commercial model sharing" copy guardrail | Owner/legal | A10 | Legal/provenance |
GATED | - |

109 | A12 | Reopen or defer P10/P11 corpus flywheel: `PATCH /api/segments`, collector
promotion, `train|reeval` job, and delta-F1 surface | `khelsutra` + `badminton-highlight-
indexer` + collector | A1, A3 | Owner D13 reopen | GATED | - |

110 | A13 | Alpha launch review: runbook, privacy/consent copy, known limits,
rollback/disable steps, and owner GO/NO-GO | Cross-repo | A1, A4, A10, A11 | Owner launch
decision | TODO | - |

111

112 ## 8. Concrete next PR sequence

```

113
114     1. **Docs PR:** land this tracker.
115     2. **Corpus portability PR:** rebase/portable manifest tooling plus missing-file smoke
test.
116     3. **Hosted alpha PR/batch:** health/auth/upload/process/job/compile/download
verification and runbook updates.
117     4. **Baseline PR:** intentional n=15 nightly baseline refresh and heuristic floor
refresh.
118     5. **Ablation/fusion PRs:** fix ablation import, complete 15-video features, rerun
promotion/rejection decisions.
119     6. **Launch decision PR/docs update:** record ship-gate target, C3 status, alpha copy
limits, and launch checklist.
120
121     ## 9. Open questions
122
123     1. **Owner:** Is the first alpha allowed to use manual/offline corpus promotion, or
should P10/P11 be reopened before any tester wave?
124     2. **Owner/platform:** Which hosted compute target is the next stand-up: Cloud Run GPU,
GCE L4 VM, or another approved GPU host?
125     3. **Owner/legal:** What is the acceptable alpha wording while WASB C3 remains
unresolved?
126     4. **Engineering:** Should the portable corpus manifest live as a generated artifact, a
checked-in redacted manifest, or both?
127     5. **Engineering:** Should n=15 feature regeneration target full fusion schema for every
clip, or should the eval loaders support mixed schemas with explicit capability tags?
128
129     ## 10. Definition of done
130
131     - [ ] Invited tester can complete the hosted record/upload -> reel download CUJ without
SSH.
132     - [ ] Alpha service is gated behind the chosen auth posture and has a documented
disable/rollback path.
133     - [ ] n=15 corpus is portable enough that manifest-driven quality commands do not rely
on stale local absolute paths.
134     - [ ] n=15 nightly baseline and heuristic floor are refreshed with recorded commands.
135     - [ ] R2/tolF1 gate is promoted or rejected with a recorded ablation result.
136     - [ ] Fusion/person/Gate-A served-default decisions are explicitly promoted or rejected
on the 15-video corpus.
137     - [ ] Ship-gate target and allowed alpha claims are recorded.
138     - [ ] WASB C3 status is recorded and reflected in alpha messaging.
139     - [ ] P10/P11 are either reopened with a committed sequence or explicitly deferred for
alpha-plus.
140     - [ ] Owner records GO/NO-GO for the first alpha wave.
141
142     ## 11. References
143
144     Internal docs:
145
146     - `docs/NEXT_STEPS.md`
147     - `docs/ROADMAP.md`
148     - `docs/data_pipeline/GOLDEN_VIDEOS.md`
149     - `docs/R2_EVAL_METRIC_DESIGN.md`
150     - `docs/RALLY_DETECTION_FIXES_PLAN.md`
151     - `docs/COMPUTE_DECOUPLED_SERVING/README.md`
152     - sibling `khehsutra/NEXT_STEPS.md`
153     - sibling `khehsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`
154     - sibling `khehsutra/docs/PER_RALLY_SELECTION.md`
155     - sibling `sports-data-collector/data/golden_source_videos.json`
156
157     Local verification commands used for the 2026-07-03 snapshot:
158
159     ```powershell
160     python -m backend.eval.fusion_compare --features-dir output
161     python -m backend.eval.serve_contrast --manifest <rebased-n15-manifest>
162     python -m training.gen0.harness --manifest <rebased-n15-manifest> --floor

```

```

eval_baselines\heuristic_lovo_n6.json --version analysis-n15 --out <temp-out>
163     python scripts\nightly_regression.py --manifest <rebased-n15-manifest> --features-dir
output --no-report
164     python -m backend.eval.ablation --features-dir output --granularity group
165     ```
166
167     ---
168
169     ### Changelog
170
171     - 2026-07-03 - Created from a synced five-repo review of `NEXT_STEPS`, `ROADMAP`,
golden-corpus docs, local eval artifacts, and the current 15-video corpus state.
172

```

6. assistant (2026-07-03T16:39:49.749Z)

Now let me check the updated README.md and other files to verify the markdown links and cross-references:

7. user (2026-07-03T16:39:49.992Z)

```

1     - [RALLY_QUALITY_RESEARCH.md](RALLY_QUALITY_RESEARCH.md) ? Foundation doc for
closing the local-vs-Gemini rally-detection gap: the measured gap + the code-grounded over-merge
diagnosis ("shuttle moving ? rally"), an inventory of the eval/experiment harness + its gaps,
the strengthened evaluation framework (segmental Edit/F1@k that make over-segmentation first-
class, golden-corpus + active learning, Gemini-agreement as a shadow signal), an
adversarially-verified, cited external-research synthesis (TAL/TAS boundary heads, racket-
sports rally-state cues, small-data, eval rigor), a prioritized cheap?durable roadmap with
falsifiable success criteria, and (§7) a tracked project plan ? work-item checklist,
sequencing, the single quality number attributed to the effort, and a definition of done
(project completes when all tiers land + the number is recorded). The plan that the
[QUALITY_ITERATIONS.md](QUALITY_ITERATIONS.md) log executes against; ties together
EXPERIMENT_HARNESS / MULTI_SIGNAL_FUSION_PLAN / ANALYZERS_AND_RECONCILERS /
OWNED_MODEL_IMPLEMENTATION_PLAN.
2     - [RALLY_DETECTION_GAP_CLOSURE.md](RALLY_DETECTION_GAP_CLOSURE.md) ? Tracked execution
doc (2026-06-18) for closing the remaining rally-detection accuracy gap per video, before
launch, driven by the growing golden corpus. First ground-truth clip (GX010141) reframed it:
boundaries are tight (R2/R3/R4 holds) ? the gap is detection: (G1) recall (local & Gemini
both miss ~half on hard clips, and different rallies ? complementary), (G2) precision (local
over-fires), (G3) rally-END over-extension on the quick-return-after-point (shuttle still
moving ? needs a player-state signal). Levers (owner-gated): player-kinematics + shuttle-in-
hand / sustained-invisibility end-rule with reverse-timeline backtrack, rally_gate Gate A
wiring, complementary fusion. Peer of RALLY_QUALITY_RESEARCH.md (execution tracker, not
research); §7 is the source of truth.
3     - [PROMINENT_COURT_DETECTION.md](PROMINENT_COURT_DETECTION.md) ? Design + experiment
(2026-06-19, default-OFF, owner-gated flip) for the multicourt G2 over-fire: identify the
prominent (foreground) court (the one covering the majority of the frame) and keep only
rallies whose shuttle lives inside it ? re-defining the shuttle signal (x,y,visible) ?
(x,y,visible-AND-inside-prominent-court) before windowing, so a background/adjacent-court
shuttle never forms a rally. 2D floor polygon (MVP) ? pseudo-3D play-volume (refinement, so
high clears aren't clipped). Born from the GX010142 rally-#1 background-court FP; reuses
fusion_features.point_in_polygon + FusionConfig.court_polygon. §8 progress tracker is the
source of truth.
4     - [RALLY_DETECTION_QUALITY_REPORT.md](RALLY_DETECTION_QUALITY_REPORT.md) ? Technical-
paper checkpoint (2026-06-17) of rally-detection quality: the honest measured state (over-
segmentation milestone tolerance-F1 0.091?0.323 at n=11, p=0.002; merge-rate 0.694?0.150; first
real-footage ground-truth at-par with START at Gemini parity), the task-faithful R2 metric, a
detailed Gemini comparison, next steps + expected wins, areas yet to be explored, and a
publishability / path-to-submission verdict. Renders to a polished PDF; the externally-shareable
"toward a paper" summary of RALLY_QUALITY_RESEARCH.md + QUALITY_ITERATIONS.md.
5     - [EVALUATION.md](EVALUATION.md) ? How we measure and improve quality.
6     - [EXPERIMENT_HARNESS.md](EXPERIMENT_HARNESS.md) ? Design (P1 SHIPPED) for A/B + shadow
+ SxS experimentation on rally-detection variants with zero production regression ?
experiments are decoupled from deploy (new cues default OFF, promote only on measured LOVO lift,
rollback = flip a flag). Composes the existing backend/eval/ machinery (calibration,

```

```

`regression`, `metrics`); implemented in `backend/eval/experiment.py`.
7      - [QUALITY_ITERATIONS.md](QUALITY_ITERATIONS.md) ? Living log of rally-detection
quality iterations (hypothesis ? what shipped ? measured/expected effect ? lesson). The reverse-
chronological record; terminal-state snapshots get archived under `archives/quality-
iterations/`.
8      - [ANALYZERS_AND_RECONCILERS.md](ANALYZERS_AND_RECONCILERS.md) ? Design (owner-gated,
NOT started): a signal-fusion framework for rally detection ? producers emit confidence-
scored signals at three temporal scopes (frame detectors / window analyzers / session
analyzers), a learned scorer weights them, and an auditable report-first reconciliation
pass lints the decision against the evidence. The spine: no special-casing in the decision
path (signals?scorer, never `if/else`). Worked examples: a service-fault analyzer, a warm-
up/practice span detector, and consuming a per-frame shuttle-state detector. A peer to
`EXPERIMENT_HARNESS.md` and `MULTI_SIGNAL_FUSION_PLAN.md` (ADR-007 modular rally layer).
9      - [PERSONALIZATION_PLAN.md](PERSONALIZATION_PLAN.md) ? Design (owner-adopted hybrid;
Phase-0 landing) for the per-user personalization feature on a generalization-first
baseline: a thin per-user tuning layer gated by the honest small-N statistics (a min-N
promotion floor + a structural-guard exemption, never a service gate) + a contribution
flywheel (free-tier videos pool into the owner-trained baseline; "a tool that gets better the
more you help it"). Records the locked owner decisions (identity, storage, min_n
semantics, Gate-A default, free-tier pooling). Built so far: per-user promotion gate (#141) +
held-out-USER learning curve (#142).
10
11     - [GOLDEN_REGRESSION_FIXTURES.md](GOLDEN_REGRESSION_FIXTURES.md) ? Design (owner-
gated, NOT started): video-free, non-attributable regression fixtures so a clean clone runs
the rally-seg suite with no video/GPU/DB. Two layers ? the post-detector freeze + dual
gates (characterization + quality-floor) reusing the `backend/eval/` stack, and the SEALED-
FIXTURES privacy filter (de-attribution-at-source, tiered public/key-gated, shuttle-only
public tier). Carries a cited IN/US/EU legal memo (derived-data IP/privacy), a threat model,
and a $7 progress tracker + definition-of-done. Extends
[GOLDEN_DATA_SHARING.md](data_pipeline/GOLDEN_DATA_SHARING.md).
12
13     ## ? Documentation status & governance
14     - [AUDIT_REPORT_khelsutra-guru_2026-07-02 (? COMPLETE /
ARCHIVED)](archives/past_projects/AUDIT_REPORT_khelsutra-
guru_2026-07-02-COMPLETED-2026-07-02.md) ? the five-repo `khelsutra-guru` workspace code audit +
its folded-in Remediation tracker (P0?P28 all closed). Archived 2026-07-02 once every
remediation wave landed and the last residuals were closed or filed as issues (#443 CORS, #444
corpus Phase-2, #445 GCS eval; + #301/#327/#332/#384). Absorbed the prior
`CODE_CLEANUP_AUDIT_2026-06-24` + June-2026 hardening sweep (both archived).
15     - [DOC_STATUS.md](DOC_STATUS.md) ? living Documentation Status & Archival
Register: every foundational/tracked doc's lifecycle + archive-readiness, the archival due-
diligence checklist ("honestly update ? then archive"), and the open doc-hygiene findings.
Review here; update rows as docs complete.
16
17     ## Product / platform plans
18     - [ALPHA_LAUNCH_READINESS.md](ALPHA_LAUNCH_READINESS.md) ? Active tracked project
(`DRAFT`, created 2026-07-03): cross-repo alpha readiness tracker after the golden corpus
reached 15 videos. Separates the hosted product-alpha lane (shareable upload?process?pick?reel)
from the owned-model/commercial rally-detection gate (ship target + WASB-C3 still unresolved).
$7 is the source of truth for the concrete next PR sequence: corpus portability, n=15 baseline
refresh, ablation/fusion reruns, hosted serving, alpha copy/launch review.
19     - [COMPUTE_DECOUPLED_SERVING/](COMPUTE_DECOUPLED_SERVING/README.md) ? Active tracked
project (2026-06-21, `DRAFT` owner-gated): the productionization successor to
DECOUPLED_COMPUTE (its deferred M3/M4). One pure core (DETECT?WINDOW?STITCH) that never
branches on deployment profile, with a ComputeBackend (where it runs) + StorageBackend
(what bytes) selected by one startup config. Ships truly-local (local/USB + local GPU,
in-process stitch, zero cloud) + cloud-serving (upload<2GB / gdrive / bucket ? a Cloud Run
GPU job runs detect AND stitch, metered into a per-job cost ledger) + cpu-mock (CI).
Includes [TESTING_STRATEGY.md](COMPUTE_DECOUPLED_SERVING/TESTING_STRATEGY.md) (profile-PARITY
proof for $0), [architecture.svg](COMPUTE_DECOUPLED_SERVING/architecture.svg), a
[runlogs/](COMPUTE_DECOUPLED_SERVING/runlogs/) posterity dir, and ADR-014 ([lineage
ADR-009?010?011?012?013?014](archives/decisions/DECISIONS.md)). $7 M0?M10 tracker is the source
of truth.
20     - [DECOUPLED_COMPUTE/](archives/past_projects/DECOUPLED_COMPUTE/README.md) ? ?
DELIVERED + ARCHIVED (2026-06-21). Lifted the rally pipeline onto a rented cloud-native

```

GPU** doing bucket-driven `train` + `infer`: storage/compute seams (PRs #246?#259) + the **M3.5 native WASB runner**;

M0?M2 + M3.5 proven on a GCP L4 (M2 ? `weights/0.1.0-l4/`). Delivered on **GCP** ? DigitalOcean dropped for compute ([ADR-013](archives/decisions/DECISIONS.md)).

M3/M4 (serving endpoint + per-video billing + scale-to-zero) deferred ? the "Cloud Run + GPU serving" subproject ([ADR-012](archives/decisions/DECISIONS.md)). Snapshot under `docs/archives/past\_projects/DECOUPLED\_COMPUTE/` (incl. `DEPLOY\_REPORT\_GCP\_2026-06-20.md`).

21 - PACKAGING_PLAN.md ? **? Active tracked project (`IN PROGRESS`, 11 PRs merged 2026-06-21/22):** ship the engine as a **downloadable batteries-included Python app** ? `pip install` ? `indexer --app` boots a local webserver and opens the **bundled KhelSutra SPA** single-origin, **torch-free** (LIGHT tier), cross-platform, no native component (decision D10/01). Built ON the decoupled-compute seam. **Supersedes the `DESKTOP\_APP\_PLAN.md` narrative.**

\$7 tracker = source of truth (P0a?e ?, P1a/b ?, P2-launcher/P2a/P2d/P2e ?). **? Next:** P3 GPU validation + screencast ? P2b-install ? P2c wizard (cross-repo) ? P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` / `[[next-session-packaging-app]]`.

22 - DESKTOP_APP_PLAN.md ? **? Superseded narrative (by `PACKAGING\_PLAN.md`; D10 = self-hosted webserver, NOT a desktop app).** Original scoping doc (2026-06-18, `[design]`): package the local pipeline as a **cross-platform desktop app** for non-technical players ? plug in the camera's USB/SD card, click once, get back **only the highlight reels**, fully on-device (no upload, no original-video bloat). The centerpiece is the **de-WSL native-compute port** (run WASB natively per-OS: Linux/Windows CUDA · macOS MPS/CPU ? drop WSL2), and its **make-or-break unknown**: is CPU acceptable on a typical *no-GPU* enthusiast laptop (WASB is ~3.4× real-time on a laptop 2070S)?

\$7 tracker = **P0 compute-feasibility spike ? P1 native detector ? P2 app shell ? P3 packaging/installers**. Realizes the **L0 fully-local** tier of VIDEO_LOCALITY_MODEL.md; is the **`LocalDesktop` ComputeBackend** of PLATFORM_ARCHITECTURE.md S3.2; the smaller/export-clean owned model (OWNED_MODEL_IMPLEMENTATION_PLAN.md) is one answer to the compute crux.

23

24 ## Practical / ongoing

25 - CREATING_REELS.md ? **? End-user guide:** install the LIGHT-tier app ? `indexer --app` ? process a video ? pick rallies ? download a watermarked reel. Task-first usage walkthrough (web UI + the API chain), with a \$6 pointer to the cloud compute option. LIGHT tier only (offline `motion` / optional Gemini; no on-device GPU or one-click cloud). Install mechanics ? root [README.md](../README.md); packaging roadmap ? PACKAGING_PLAN.md.

26 - CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md ? **? Operator runbook:** `gs://` bucket videos ? highlight reels on the approved **L4 Cloud Run** (scale-to-zero, \$0 idle). Copy-pasteable: build image ? create the GPU Job ? cloud-serving control plane ? process(cloud)?query?compile?download ? prove the path (`verify\_compute\_path.py`) ? \$0 teardown. **Verified e2e 2026-06-23.** Raw deploy knobs ? ../deploy/cloudrun/README.md.

27 - SETUP_NEW_MACHINE.md ? **? The canonical "spin up a box ? ready for training & inference" runbook.** Copy-pasteable, human-followable, **verified end-to-end on DigitalOcean (2026-06-20)** ? but **compute is now GCP-only (ADR-013)**, see [deploy/gcp/README.md](../deploy/gcp/README.md): credentials ? provision ? bootstrap ? suite (922 passed) ? secret-free CPU inference (no GPU/keys) ? real Gemini inference ? Gen-0 training, with pointers to the GPU/native-detector + bucket paths. The `deploy/digitalocean/\*` scripts are provider-agnostic and reused on GCP.

28 - DATA_IN_GCS.md ? **? The corpus source-of-truth map: where every piece of data lives in `gs://khelsutra-rally-corpus`** (canonical layout · current contents + drift · how `backend.worker` consumes it · the still-local-only **gap** · migration plan). Read this so a session pulls data from the bucket, not from someone's `C:/F:/Drive` ? the goal is to remove the local box as a blocker. Pairs with the two runbooks below.

29 - [deploy/gcp/nightly_regression/README.md](../deploy/gcp/nightly_regression/README.md) ? **? Nightly engine-regression OPERATIONS MANUAL (one-stop shop):** enable · run on demand · change any setting (email recipients/frequency/clips/config) · observe · pause/disable/remove. The full WASB-pipeline reel-count+cost+email nightly on an ephemeral GCP GPU VM (design/status: NIGHTLY_ENGINE_REGRESSION.md). The lighter CPU cached-trajectory tripwire is `scripts/nightly\_regression.py` (local Windows Scheduled Task, PR #202).

30 - TRACKNET_WSL_SETUP.md

31 - Evaluation harness and calibration drivers (in `backend/eval/`)

32

```

33     ## Data layer (collection · annotation · golden sets ? **no ML**)
34     - [data_pipeline/](data_pipeline/README.md) ? the isolated data pillar: how
footage is recorded (VIDEO_RECORDING_GUIDELINES), golden sets are created + which videos
(GOLDEN_SET_IMPLEMENTATION · GOLDEN_SET_PHASE1_VIDEOS · GOLDEN_VIDEOS), and data is
shared/ingested (GOLDEN_DATA_SHARING). Deliberately no segmentation/stitching/model detail
? that's the ML pillar (top level + COMPUTE_DECOUPLED_SERVING/). The annotation-vendor
(Roora) engagement is archived under archives/roora-vendor/.
35
36     ## Conventions & templates
37     - [PROJECT_DOC_TEMPLATE.md](PROJECT_DOC_TEMPLATE.md) ? Reusable skeleton for a tracked
project doc: status header · decisions-locked · progress tracker (??/?) · definition-of-done
· archived on completion. Standardizes the emergent pattern in RALLY_QUALITY_RESEARCH.md §7
and GOLDEN_REGRESSION_FIXTURES.md. Use for any substantial design/project work.
38
39     ## Historical documents
40     Completed plans, old research, ADRs, and checkpoints live in [archives/](archives/).
41
42     - June-2026 test-hardening sweep ? COMPLETED (2026-06-14; T1?T5 + full loop +
verification per #162) and archived; the former top-level pointer-stubs were removed
2026-07-02 (absorbed into the umbrella's Prior audits). Full records:
[archives/CODE_AUDIT_AND_TEST_HARDENING-
COMPLETED-2026-06-14.md](archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md),
[archives/DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md](archives/DEFERRED_HARDENING_PLAN-
COMPLETED-2026-06-14.md), [archives/HARDENING_LOOP_HANDOFF-
COMPLETED-2026-06-14.md](archives/HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md).
43     - CODE_CLEANUP_AUDIT (2026-06-24) ? the prior deep engine audit (35 findings; Phase
1 complete, Phase 2 largely shipped, remaining = the main.py god-module split). Absorbed
into the umbrella (see its Prior audits (absorbed)) and archived ? [archives/past_projects/
CODE_CLEANUP_AUDIT_2026-06-24-SUPERSEDED-2026-07-
02.md](archives/past_projects/CODE_CLEANUP_AUDIT_2026-06-24-SUPERSEDED-2026-07-02.md).
44     - KHELSUTRA_GURU_HEALTH_REMEDIATION_PLAN ? the five-repo audit execution board,
folded into the umbrella as its Remediation tracker (live) and archived ?
[archives/past_projects/KHELSUTRA_GURU_HEALTH_REMEDIATION_PLAN-
FOLDED-2026-07-02.md](archives/past_projects/KHELSUTRA_GURU_HEALTH_REMEDIATION_PLAN-
FOLDED-2026-07-02.md).
45     - [DECISION_SNAPSHOT_REGRESSION.md](DECISION_SNAPSHOT_REGRESSION.md) ? Tracked
project (decisions D1?D11 LOCKED 2026-06-25, #383; implementation P1?P7 pending): a
deterministic quality-neutral proof ? freeze + compare the pipeline's decision output
(rally segments + stitch/concat plan) off a cached trajectory, GPU-free, in CI. The behavioral
proof that lets a genuine (non-AST-identical) refactor be confidently called quality-neutral.
Never hashes the encoded reel (non-deterministic). Extends the existing characterization
snapshot (test_golden_fixtures.py) from segmentation to the stitch plan; peer of the M4
quality-floor + GPU nightly_reelcount.py. §7 tracker is the source of truth.
46
47     - Field-ops / customer-feedback (recruiting flyer · field-visit playbook · PMF) ?
[archives/field-pmf/](archives/field-pmf/) (archived 2026-06-21; PMF validation still an
active business thread ? see NEXT_STEPS.md).
48     - Roora annotation-vendor engagement (labeling brief · vendor bake-off) ?
[archives/roora-vendor/](archives/roora-vendor/) (archived 2026-06-21).
49     - Superseded: per-video output layout ? [archives/PER_VIDEO_OUTPUT_LAYOUT-
SUPERSEDED-2026-06-21.md](archives/PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md) (?
PER_VIDEO_WORKSPACE.md).
50
51     See archives/README.md for the umbrella policy and archives/past_projects/README.md
for classification of small tech debt / starter projects.
52

```

8. user (2026-07-03T16:39:50.055Z)

```

1     # Documentation Status & Archival Register
2
3     > Status: ? LIVING ? the single board for doc lifecycle + archival due-
diligence. Last full review: 2026-07-02.
4     > Update cadence: whenever a doc changes lifecycle state (created / shipped a
milestone / delivered / archived), and re-review at each clean-slate.

```

```

5      > **Why this exists:** so we can (a) **review completeness** at a glance, (b) **update
it as work finishes** to prove we're done, and (c) **never archive a foundational doc without
honestly updating it first** (owner directive, 2026-06-21).
6
7      ---
8
9      ## 0. How to use this
10     - **Reviewing now?** Read §4 (the open findings/actions) ? that's what needs attention.
§3 is the full register.
11     - **Marking something complete?** Update the doc's own **§7 progress tracker** (the
source of truth for its completeness), then flip its row here, then ? if terminal ? run the §2
archival checklist.
12     - **This register does NOT duplicate completeness detail** ? it indexes lifecycle +
archive-readiness and points to each tracked doc's §7.
13
14     ## 1. Lifecycle legend
15     `DRAFT ? IN PROGRESS ? DONE/DELIVERED ? ARCHIVED`. Plus: **LIVING** (permanent
reference, never "done"), **PROPOSAL/DESIGN** (owner-gated, not started), **REPORT** (a terminal
record of a run/launch), **CONVENTION** (a standing rule/guide).
16
17     ## 2. Archival due-diligence checklist ? the repeatable process (do NOT just `git
mv`)*
18     Before a foundational/tracked doc moves to `docs/archives/`:
19     1. **VERIFY claims against the code** ? no stale assertions; tag load-bearing facts
`[verified]`. (A doc that says "nothing built" when code shipped is not archive-ready ? fix it
first.)
20     2. **Flip Status ? DELIVERED / DONE / REJECTED + a completion note** ? what shipped,
where (PRs, paths, artifacts), and what was **deferred / carried forward**.
21     3. **Update ALL inbound references** ? `docs/README.md` index, `CLAUDE.md`,
`docs/NEXT_STEPS.md`, and any cross-doc links ? the new path + status.
22     4. **Keep the lineage** ? ADR forward/back-links + "supersedes / superseded-by" so the
evolution stays traceable.
23     5. **git mv` the whole dir/file ? `docs/archives/past_projects/`** (terminal-state
ONLY; live docs stay at `docs/` top level). Relative links inside become historical (note this
in the doc's header).
24     6. **Record the move** ? flip the row here to §3e + a line in memory `current-state`.
25
26     > **Worked example (the template):** **DECOUPLED_COMPUTE ? #270** ? verified M0?M2+M3.5
delivered on GCP, flipped ??? DELIVERED with a completion note (incl. carried-forward
WASB-C3/DPA/`md5`), reindexed `docs/README` + `past_projects/README`, kept the ADR-010?014
lineage, then `git mv` to `docs/archives/past_projects/DECOUPLED_COMPUTE/`.
27
28     ## 3. The register
29
30     ### 3a. Tracked projects ? lifecycle docs (archive when DONE; each doc's §7 is the
completeness source-of-truth)
31     | Doc | Status | Archive-ready? | Next action |
32     |---|---|---|---|
33     | `ALPHA_LAUNCH_READINESS.md` | **DRAFT** ? cross-repo alpha readiness tracker created
2026-07-03 after the golden corpus reached n=15; separates hosted product-alpha work from owned-
model/commercial rally-detection gates | No | Execute §7 in order: corpus portability + n=15
baselines, hosted alpha serving, ablation/fusion reruns, then launch G0/N0-G0 |
34     | `COMPUTE_DECOUPLED_SERVING/` | **IN PROGRESS** ? M0 approved + **M1?M9 LARGELY
**SHIPPED** (#279?#290; M5 #286, M6 #287, M8 #288, M9-auth #290 all merged, default-OFF); Gen-0
weights pushed (`weights/0.1.0-l4/`) | No | M11 flywheel + M12 gdrive-api (owner residual:
counsel/prices/OAuth/real-run within the D17 cap) |
35     | `RALLY_QUALITY_RESEARCH.md` | Foundation, **tracked & ACTIVE** (§7) | No | Drives the
R-series; complete when §7.1 single quality number is attributed |
36     | `R2_EVAL_METRIC_DESIGN.md` | IMPLEMENTED (augmenting) | No (gates R4 + dual-eval;
ablation pending) | Promote-or-reject tolF1 as the gate (ablation) ? then archive with the
R-cluster |
37     | `R1_MILESTONE_LAUNCH_REPORT.md` | LAUNCHED (#204) ? terminal record | Soon (with the
R-cluster) | Keep with R-series for narrative; archive when the track wraps |
38     | `RALLY_DETECTION_GAP_CLOSURE.md` | DRAFT (owner-gated) | No | G1?G3 levers; part of
the R/quality cluster |

```


39 | `RALLY\_DETECTION\_FIXES\_PLAN.md` | ****DONE**** ? findings A/B/C resolved; P2b promoted
`inactivity\_temporal\_density` default-ON (#396) | With the R-cluster (deferred per its header;
gate = R2 toIF1 ablation + \$7.1 number) | Archive with the R-series when the track wraps |

40 | `RALLY\_DETECTION\_CODE\_REVIEW.md` | Terminal snapshot ? findings A (#394), B (#394 ?
promoted #396), C (`sdc#50`) all resolved | With the R-cluster | Archive with the R-series |

41 | `PER\_VIDEO\_WORKSPACE.md` | DRAFT, IN PROGRESS (P0 = #264) | No | ? supersede-reconcile
(finding #1) DONE; P3 de-hardcode base ? via #282; remaining P1?P6 (P1 v6 migration coordinates
with M8's v6 cost-ledger ? version bump authored once) |

42 | `MULTI\_SIGNAL\_FUSION\_PLAN.md` | P1 DONE; P2+ owner-gated | No | P3/P4 (learned fusion,
audio) ? needs GPU golden features |

43 | `EXPERIMENT\_HARNESS.md` | P1 IMPLEMENTED | No | A/B + promotion gate; supports the
quality track |

44 | `PROMINENT\_COURT\_DETECTION.md` | DRAFT (owner-gated) | No | C-series; multi-track
selection design |

45 | `EXECUTION\_POLICY.md` | P1 SHIPPED; P3b next | No | Async job policy; reused by Cloud
Run dispatch |

46 | `GOLDEN\_REGRESSION\_FIXTURES.md` | IN PROGRESS (owner-gated) | No | Golden eval
fixtures |

47 | ****Data layer**** ? `docs/data\_pipeline/` (`GOLDEN\_SET\_IMPLEMENTATION` .
`GOLDEN\_SET\_PHASE1\_VIDEOS` . `GOLDEN\_VIDEOS` . `VIDEO\_RECORDING\_GUIDELINES` .
`GOLDEN\_DATA\_SHARING`) | LIVING (data, ****no-ML****) | n/a | Relocated 2026-06-21 to the isolated
data pillar; tracked via `data\_pipeline/README.md`. Roora **vendor** docs archived ?
`archives/roora-vendor/`. |

48 | `archives/past_projects/AUDIT_REPORT_khelsutra-
guru_2026-07-02-COMPLETED-2026-07-02.md` **** (umbrella) **** | ***** COMPLETE / ARCHIVED (2026-07-02) ****
? the five-repo code audit + folded-in **Remediation tracker** (P0?P28 all closed) + absorbed
prior audits. All non-owner-gated residuals landed; owner-gated/future items filed as issues
(#443/#444/#445, + #301/#327/#332/#384). | Archived | Owner confirmed archival 2026-07-02; moved
to `archives/past\_projects/` per §2. Residuals now live as GitHub issues, not in this doc. |

49 | `DECISION\_SNAPSHOT\_REGRESSION.md` | ****DRAFT ? decisions LOCKED**** (D1?D11 locked
2026-06-25, #383). Zero code shipped for P1?P7. | No | Start P1+P2 (schema + stitch-plan
factor); de-risk the P10 main.py split |

50 | `DESKTOP\_APP\_PLAN.md` | DRAFT (owner-gated, nothing built) | No | P0 compute-
feasibility spike; relates to truly-local profile |

51 | `OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md` | PLAN/roadmap (owner-gated) | No (authoritative
roadmap) | Gen-0?Gen-2; the owned-model track |

52 | `PERSONALIZATION\_PLAN.md` | DESIGN ? owner-adopted; Phase-0 | No | Per-user models (DB
v5 store landed) |

53 | `UI\_ALPHA\_EXECUTION\_PLAN.md` | ACTIVE | No | khelsutra UI alpha track |

54 | `I18N\_PLAN.md` | Design (P0); ****in implementation**** (adapted) | No | Executed in
`khelsutra/docs/FRIENDLY\_AND\_MULTILINGUAL.md` (P1?site/, P3?web/); P2 backend stays here, owner-
gated |

55 | `POST\_166...RISK\_ASSESSMENT.md` | Plan for owner review | No | Milestones/risk doc |

56

57 **### 3b. Foundational / strategy references (LIVING ? archive only if ****superseded****, not
when "done")**

58 | Doc | Status | Note |

59 | ---|---|---|

60 | `PLATFORM\_ARCHITECTURE.md` | Strategy/research | The `ComputeBackend`/`StorageBackend`
registries ? realized by COMPUTE_DECOUPLED_SERVING |

61 | `COMMERCIALIZATION.md` | ? LIVING | Strategy; carries WASB-C3 |

62 | `VIDEO\_LOCALITY\_MODEL.md` | THINKING DOC | L0 fully-local = the truly-local profile |

63 | `HOSTING\_PLAN.md` | PROPOSAL | ******* partly overtaken by ADR-013 + the shipped site ?
verify (#5)****** |

64 | `UI\_PROPOSAL.md` | PROPOSAL (partly shipped) | UX half ****in implementation**** ?
`khelsutra/docs/FRIENDLY\_AND\_MULTILINGUAL.md`; currency banner added (finding #5 ? for this doc)
|

65 | `STORAGE\_SHARING\_MODEL.md` | DRAFT (co-design pending) | ? |

66 | `ANALYZERS\_AND\_RECONCILERS.md` | DESIGN (owner-gated) | ? |

67 | `ROADMAP.md` / `CODE\_MAP.md` / `HOW\_RALLY\_DETECTION\_WORKS.md` /
`KHELSUTRA\_PRODUCT\_OVERVIEW.md` | LIVING references | Keep current; not archive targets |

68

69 **### 3c. Reports & terminal records (archive candidates once their track wraps)**

70 | Doc | Status | Note |

71 | ---|---|---|

72 | `REAL\_FOOTAGE\_VALIDATION\_REPORT.md` | validation complete | Motivates R2; archive with the R-cluster |

73 | `RALLY\_DETECTION\_QUALITY\_REPORT.md` | report | Quality snapshot; part of R-cluster |

74 | `QUALITY\_ITERATIONS.md` | LIVING empirical log | ****Keep**** ? the paper-backbone ablation log ([[paper-coauthor-aim]]) |

75 | `CODE\_AUDIT\_AND\_TEST\_HARDENING` · `HARDENING\_LOOP\_HANDOFF` · `DEFERRED\_HARDENING\_PLAN` | COMPLETED ? ****finish-archived 2026-07-02**** (top-level stubs removed; absorbed into the umbrella's ***Prior audits***) | ? Full copies under `docs/archives/\*-COMPLETED-2026-06-14.md`; inbound links repointed (finding #4 resolved) |

76

77 **### 3d. Conventions / guidelines / templates / runbooks (permanent ? do not archive)**

78 `PROJECT\_DOC\_TEMPLATE.md` · `NEXT\_STEPS.md` (? refreshed 2026-06-21) · `BACKLOG.md` · `EVALUATION.md` · `ABLATION\_LAYER.md` · `SETUP\_NEW\_MACHINE.md` · ****CREATING_REELS.md**** (? end-user usage guide ? install ? process ? pick ? reel; LIGHT tier; added 2026-06-23) · ****CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md**** (? operator runbook ? `gs://` videos ? reels on L4 Cloud Run; verified e2e 2026-06-23) · `TRACKNET\_WSL\_SETUP.md` · `README.md`. ***(2026-06-21 relocation: the data-layer guides moved ? `data\_pipeline/`; the field-ops + Roora docs ? `archives/{field-pmf,roora-vendor}/`.)***

79

80 **### 3e. Already archived (`docs/archives/`)**

81 `past\_projects/DECOUPLED\_COMPUTE/` (? 2026-06-21, #270) · `past\_projects/low-regret-moves/` · ****field-pmf/**** (field-ops + customer-feedback, 2026-06-21) · ****roora-vendor/**** (annotation-vendor engagement, 2026-06-21) · `PER\_VIDEO\_OUTPUT\_LAYOUT-SUPERSEDED-2026-06-21.md` · `CODE\_AUDIT\_AND\_TEST\_HARDENING-COMPLETED-2026-06-14.md` · `DEFERRED\_HARDENING\_PLAN-COMPLETED-2026-06-14.md` · `HARDENING\_LOOP\_HANDOFF-COMPLETED-2026-06-14.md` · `COMMERCIAL\_READINESS\_REVIEW.md` · `MONETIZATION\_AUDIT.md` · `TEST\_RUN\_SUMMARY.md` · `research/` · `checkpoints/` · `quality-iterations/` · `past\_projects/CODE\_CLEANUP\_AUDIT\_2026-06-24-SUPERSEDED-2026-07-02.md` (absorbed ? umbrella) · `past\_projects/KHELSUTRA\_GURU\_HEALTH\_REMEDIATION\_PLAN-FOLDED-2026-07-02.md` (folded ? umbrella).

82

83 **### 3f. ADR log ? `docs/archives/decisions/DECISIONS.md`**

84 ****ADR-001 ? ADR-014**** (14, all ratified). The compute/serving lineage: ****ADR-005 ? 008 ? 009 ? 010 ? 011 ? 012 ? 013 ? 014****. ADR-014 was ****RATIFIED 2026-06-21**** (owner M0 sign-off for COMPUTE_DECOUPLED_SERVING).

85

86 **## 4. Open due-diligence findings / actions (2026-06-21 review)**

87

88 > ****? Doc-restructuring done (2026-06-21, owner-directed):**** field-ops/customer-feedback ? `archives/field-pmf/`, Roora annotation-vendor ? `archives/roora-vendor/`, and a no-ML ****data layer**** carved into `docs/data\_pipeline/`. Inbound + outbound links rewired; three dir READMEs added (the `data\_pipeline` one sets the "no segmentation/stitching" boundary).

89 1. ****? RESOLVED (2026-06-21) ? `PER\_VIDEO\_WORKSPACE.md` supersedes `PER\_VIDEO\_OUTPUT\_LAYOUT.md`.**** Folded forward the `video\_slug` naming option (workspace \$8.5) + the `output/GX010125\_run/` manual-prototype precedent; predecessor archived ? `docs/archives/PER\_VIDEO\_OUTPUT\_LAYOUT-SUPERSEDED-2026-06-21.md`; `CODE\_MAP.md` repointed.

90 2. ****? RESOLVED (2026-06-21) ? `NEXT\_STEPS.md` refreshed**** with a current 2026-06-21 state + a ****cross-track prioritized pending list****; the 2026-06-13 owned-model/quality body is kept (clearly marked) as that track's detail.

91 3. ****R-series / quality cluster is ACTIVE**** (`R1`, `R2`, `RALLY\_DETECTION\_GAP\_CLOSURE`, `RALLY\_QUALITY\_RESEARCH`, `QUALITY\_ITERATIONS`, `REAL\_FOOTAGE\_VALIDATION\_REPORT`, `RALLY\_DETECTION\_QUALITY\_REPORT`). ****Do NOT archive yet**** ? archive as a ***cluster*** once R2 is promoted-or-rejected as the gate + \$7.1's single number is attributed. (Owner already confirmed "leave as is", 2026-06-21.)

92 4. ****? RESOLVED (2026-07-02) ? finish-archived the hardening stubs.**** The three top-level pointer-stubs (`CODE\_AUDIT\_AND\_TEST\_HARDENING`, `HARDENING\_LOOP\_HANDOFF`, `DEFERRED\_HARDENING\_PLAN`) were removed; full archived copies remain under `docs/archives/`, inbound links (README, PLATFORM_ARCHITECTURE, DESKTOP_APP_PLAN) were repointed, and the sweep is summarized in the umbrella's ***Prior audits (absorbed)***.

93 5. ****HOSTING_PLAN.md / `UI\_PROPOSAL.md` are PROPOSAL but partly overtaken**** by ADR-013 (DO dropped) + the shipped khelsutra site. ****? `UI\_PROPOSAL.md` banner added (2026-06-21)**** ? its UX half is now tracked in `khelsutra/docs/FRIENDLY\_AND\_MULTILINGUAL.md`; the `I18N\_PLAN.md` localization track is likewise in implementation there. ****HOSTING_PLAN.md banner still pending**** (ADR-013 already noted in it).

94 6. ****? RESOLVED (2026-06-21) ? `COMPUTE\_DECOUPLED\_SERVING` M0 approved + M1?M4 SHIPPED**** (#279/#280/#282/#283); all \$8 gating Qs locked (D7?D17); M5+ default-OFF code batch authorized

(owner residual: counsel ToS/DPA, real GCP prices, OAuth app, real cloud run within the D17 \$100/?10k cap).

95
96 ## 5. Maintenance
97 Update a row when its doc changes lifecycle; run \$2 before any archive; re-review \$4 at each clean-slate. This register + the \$2 checklist are the operational form of the `CLAUDE.md` archive policy and [[working-agreement-tracked-project-docs]].
98

9. user (2026-07-03T16:39:50.350Z)

1 # Next steps (live) ? refreshed 2026-06-21
2
3 > **How to read:** this top section is the **current cross-track state + prioritized pending list** (refreshed 2026-06-21, replacing the stale 2026-06-13 header). The detailed **owned-model & rally-quality track** plan (2026-06-13) is **preserved below, still valid** ? it is now *one of several* active tracks. Live blow-by-blow: memory `current-state.md`. Doc lifecycle/hygiene: `docs/DOC\_STATUS.md`.
4
5 ## ? Where we are (2026-06-21)
6 **DECOUPLED_COMPUTE delivered on GCP + archived** (#270); **compute = GCP only** (ADR-013, DO dropped). **`COMPUTE\_DECOUPLED\_SERVING`** subproject spun up (#272, **ADR-014**) ? one pure core + local/cloud profiles; **M0?M9 LARGELY SHIPPED** (ADR-014 ratified 2026-06-21; M1?M4 #279?#283, M5 #286, M6 #287, M8 #288, M9-auth #290 all merged, default-OFF). **Per-video workspace P0** shipped (#264); **doc-status register + archival due-diligence** in place (#273). The **khelsutra site + per-rally UI** ship in parallel (sibling track). The **rally-quality / owned-model** track (below) is the core ML work, gated on **growing the golden corpus**.
7
8 > **? LAUNCH DECISION ? NO-GO (owner, 2026-06-30).** The rally-detection product is **not** cleared for an external/commercial launch; we stay in **development**. This is the expected default for the dev phase. Two unblockers gate a future GO: **(1)** set + meet the **ship-gate target (#172)** ? currently unset, needs corpus growth (and the R2 toF1 ablation, item 8); **(2)** resolve the **WASB shuttle-detector weight provenance** (commercial-clean licensing). P2b (#396) was a do-no-harm quality increment, **not** a launch gate ? merging dev PRs ? launching. Re-decide only when both unblockers clear.
9
10 ## ? Prioritized pending ? all tracks
11 **P0 ? highest leverage / unblocks the most**
12 0. **? [owner-priority, 2026-06-21] MAKE IT UI-DRIVEN + ALPHA-SHAREABLE ? "a few recordings lying around ? highlights, from a page, no SSH".** Today the cloud flow is **SSH/CLI-only** (`docs/CLOUD\_GPU\_DRYRUN\_GUIDE.md`); **M12 does NOT fix it.** Needs (a separate khelsutra `web/` SPA + deploy track, NOT an engine seam): **(i)** deploy the engine as a reachable **service** (M7 ? Cloud Run service, ? owner GCP spend for a *persistent* endpoint), **(ii)** the **SPA ingest/progress screen** (khelsutra `web/` B-P2: enter/upload video ? `/api/process` ? poll `/api/jobs` ? reel; ZERO engine code), **(iii)** flip **M9 edge-auth ON** (merged #290, default-OFF) to gate alpha tenants, **(iv)** a **compute PICKER in the UI** ("your machine vs our cloud", D13/D15). The engine seams make it *possible*; the SPA makes it *usable*. Full record: [[ui-driven-cloud-serving-gap]]. **Concrete execution tracker after the n=15 corpus review:** `docs/ALPHA\_LAUNCH\_READINESS.md` §7.
13 0b. **? [packaging track, IN PROGRESS ? 11 PRs merged 2026-06-21/22] Downloadable batteries-included app ? `docs/PACKAGING\_PLAN.md`. The LIGHT-tier engine is bundle-ready: `pip install` ? `indexer --app` serves the **bundled SPA** single-origin, **torch-free**, with app-home state / ffmpeg resolver / DNS-rebinding guard / single-instance lock / safe DB upgrades (P0a?e, P1a/b, P2-launcher/P2a/P2d/P2e). **? Next (fresh focused pushes):** (1) P3 **GPU validation + screencast** (L4, ~\$0.50, DELETE-after, \$100 cap ? `[[gcp-vm-always-delete]]`; ? native-WASB 0-rally tuned-config blocker), (2) P2b-install (uv/script + truly-local default config), (3) P2c first-run wizard (cross-repo khelsutra SPA from `/api/capabilities`), then P3-cloud (Cloud Run) + P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` / `[[next-session-packaging-app]]`.
14 1. ? **`COMPUTE\_DECOUPLED\_SERVING` M0?M9 LARGELY SHIPPED** (2026-06-21, ADR-014 ratified). M1?M4 (#279/#280/#282/#283), **M5 #286, M6 #287, M8 #288, M9-auth #290** all merged (default-OFF); the **real M7 L4 run** validated the worker (22 rallies, VM deleted, \$0). All \$8 gating Qs locked (D7?D17). **Owner residual** (not blocking Claude's default-OFF code): counsel ToS/DPA (M9-consent live + M11 live), real GCP prices+FX (M8 calibrate), Google OAuth app (M12 real), CF team-domain/AUD + ingress-lock; real cloud verification authorized within a

****\$100/?10k cap** (D17).**

15 2. ****[owner, ongoing] Grow the golden corpus**** ? multi-court badminton + ?3 matches/sport (TT first). The single highest-leverage move; the quality track ***and*** full-corpus training feed on it.

16 3. ****[owner + Claude + GPU] Full-corpus training (n?5)**** ? a real ``weights/<ver>/`` (the ``0.1.0-l4`` bundle is n=2 plumbing-only).

17

18 ****P1 ? near-term, mostly Claude-doable (default-OFF, green-gated)****

19 3b. ****[Claude] Five-repo audit health remediation ? ? COMPLETE (2026-07-02).**** All waves (P0?P28) landed and the last residuals were closed or filed as issues; the umbrella tracker was archived to ``docs/archives/past_projects/AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md``. Remaining owner-gated/future work lives as GitHub issues: #443 (CORS default), #444 (corpus Phase-2 upload), #445 (GCS eval plumbing), plus #301/#327/#332/#384.

20 4. ****[Claude] ``COMPUTE_DECOUPLED_SERVING`` batch ? REMAINING: M11 + M12**** (M5/M6/M8/M9-auth ? merged this session). ****M11**** = ``backend/flywheel.py`` capture?store?shadow-eval?goldenize plumbing reusing ``workspace.py::for_video`` + ``backend/eval/experiment.compare`` + ``backend/eval/promotion.py::promote_if_served_safe``, behind ``flywheel.*`=OFF` + a faked ``consent_attested`` (? privacy-sensitive ? give it fresh focus). **\*\*M12\*\*** = ``backend/storage/gdrive_api.py`` ``GDriveApiStorage(StorageBackend)`` + a ``gdrive-api://`` scheme (regex already allows the hyphen; add to ``_KNOWN`` + ``_backend_class`` + the `StorageConfig Literal`) against an INJECTED fake Drive (mirror ``gcs.py`` DI); keep the mount-only ``gdrive://`` untouched. ? ****M9-consent cols = a new v7 migration**** (M8 took v6). M9-consent ToS/DPA text = owner/counsel.

21 5. ****[Claude] Per-video workspace P1?P6**** (P1 v6 DB migration ? P2 worker ? P3 per-video routing of ``detector/+`tmp/`` `*[the `output/chunks_` de-hardcode base = ? done via #282]*` ? P4 ? P5 Launch-Report flip ? P6 GPU parity). ? P1's v6 migration coordinates with the M8 v6 cost-ledger migration ? author the version bump once.

22 6. ? ****Ops-Agent codified (#291)**** ? ``deploy/gcp/create_gpu_vm.sh`` (the one-command VM helper) always bakes the Ops-Agent startup-script; provision.sh grants the APIs+SA roles. PROVEN active on a fresh VM 2026-06-21. ([[gcp-ops-agent-enable]])

23 7. ****[owner] Set the ship-gate target (#172)**** ? needs corpus growth.

24 8. ****[owner-side GPU] R-series: promote-or-reject `R2` tolF1 as the gate**** (the ablation) ? needs GPU-extracted golden features.

25

26 ****P2 ? later / gated / deferred****

27 9. Rally-quality ****P3/P4**** (person-fusion LOVO, audio cue) ? needs GPU golden features.

28 9b. ? ****DONE (#396):**** the R4 density fix (finding B / P2b in ``RALLY_DETECTION_FIXES_PLAN.md``) is ****promoted to default-ON****. Golden-set A/B at the ****SERVED**** windowing showed tolF1 0.353?0.364 (?tolF1 +0.010, no regression), so ``inactivity_temporal_density`` default was flipped to True. (Landed flag-gated default-OFF in #394.) Fixed the sparse-tracking quirk where one fast sample after a detection gap held the rally tail open.

29 10. ****M0.4 scorer swap**** ? TemporalMaxer (per the 2026-06-17 external review).

30 11. ****Multisport**** (table-tennis first, then pickleball).

10. assistant (2026-07-03T16:39:53.123Z)

Now let me verify the specific section references and dependency links in the tracker table of Section 7. Let me carefully examine the dependencies and cross-references:

11. user (2026-07-03T16:39:54.783Z)

```
cd07339 docs: add alpha launch readiness tracker
e4914d1 Surface validation failure reasons
f0814d5 Normalize upload filenames before extension checks
e585965 Fix ffmpeg concat clip paths (#449)
ac6184b docs: archive the completed 2026-07-02 workspace audit umbrella (#446)
On branch codex/alpha-readiness-plan
Your branch is up to date with 'origin/codex/alpha-readiness-plan'.
```

nothing to commit, working tree clean

12. assistant (2026-07-03T16:39:56.844Z)

Now I'll check the specific cross-references and any potential issues with section numbers, link validity, and dependency consistency:

13. user (2026-07-03T16:39:57.843Z)

(Bash completed with no output)

14. assistant (2026-07-03T16:39:59.687Z)

Let me check for section references using a different pattern:

15. user (2026-07-03T16:40:00.708Z)

5:> ****Tracking anchors:**** Section 7 is the progress tracker and source of truth; indexed in `docs/README.md` and `docs/DOC\_STATUS.md`.

16. assistant (2026-07-03T16:40:02.662Z)

Let me systematically check the dependencies in the table at line 95-110:

17. user (2026-07-03T16:40:03.150Z)

```
91      ## 7. Deliverables & progress tracker
92
93      Legend: TODO = not started, IN PROGRESS = actively being worked, DONE =
shipped/verified, GATED = blocked on owner/legal/platform decision. One PR per row unless the
row explicitly names a cross-repo batch.
94
95      | ID | Deliverable | Repo / owner | Depends on | Gated? | Status | PR |
96      |----|-----|-----|-----|-----|-----|----|
97      | A0 | Create this alpha readiness tracker and index it in docs | `badminton-highlight-
indexer` | - | No | IN PROGRESS | #451 |
98      | A1 | Stand up alpha serving endpoint: Cloud Run/GPU or approved host, health check,
edge auth ON, upload -> process -> jobs -> compile -> download verified | `khesutra` +
`badminton-highlight-indexer` | A0 | Owner spend / CF config | TODO | - |
99      | A2 | Make the n=15 golden manifest portable/reproducible; add a path/corpus smoke
check that fails loudly when labels or trajectories are missing | `badminton-highlight-indexer`
| A0 | No | TODO | - |
100     | A3 | Promote the 15-video source-video/MD5 records into the collector/vault path;
reconcile collector's six-video source manifest with the 15-video eval corpus | `sports-data-
collector` + vault | A2 | Owner upload/storage scope | TODO | - |
101     | A4 | Refresh the n=15 nightly regression baseline intentionally and record the exact
command/output; keep stale-baseline warnings until reviewed | `badminton-highlight-indexer` | A2
| Reviewer sign-off | TODO | - |
102     | A5 | Recompute the heuristic n=15 floor and replace the stale `heuristic_lovo_n6`
floor for future Gen-0 comparisons | `badminton-highlight-indexer` | A2 | No | TODO | - |
103     | A6 | Fix `backend.eval.ablation` CLI circular import, then run the R2/tolF1 ablation
on the n=15 corpus | `badminton-highlight-indexer` | A2, A4 | No | TODO | - |
104     | A7 | Regenerate or upconvert full-fusion feature JSONs for the newer 9 videos so
fusion/person/audio analyses run over all 15 | `badminton-highlight-indexer` | A2 | GPU/data
availability | TODO | - |
105     | A8 | Re-run `fusion_compare`, group ablation, and served contrast after A7; explicitly
promote/reject person fusion, Gate A served-default, and any other R-series default flips |
`badminton-highlight-indexer` | A6, A7 | Reviewer sign-off | TODO | - |
106     | A9 | Run Gen-0/TemporalMaxer shadow training against the refreshed n=15 floor; produce
a non-serving artifact bundle and gate verdict | `badminton-highlight-indexer` | A5, A8 | Owner
approval for M0.4 scope | TODO | - |
107     | A10 | Decide and record the alpha ship-gate target: metric, corpus slice, minimum
threshold, significance rule, and what claims are allowed | Owner + `badminton-highlight-
indexer` | A4, A5, A8 | Owner decision | GATED | - |
108     | A11 | Resolve or explicitly defer WASB C3 provenance for alpha messaging; if deferred,
document "no commercial model sharing" copy guardrail | Owner/legal | A10 | Legal/provenance |
GATED | - |
109     | A12 | Reopen or defer P10/P11 corpus flywheel: `PATCH /api/segments`, collector
```

```
promotion, `train|reeval` job, and delta-F1 surface | `kheIsutra` + `badminton-highlight-indexer` + collector | A1, A3 | Owner D13 reopen | GATED | - |
110 | A13 | Alpha launch review: runbook, privacy/consent copy, known limits,
rollback/disable steps, and owner GO/NO-GO | Cross-repo | A1, A4, A10, A11 | Owner launch
decision | TODO | - |
111
112 ## 8. Concrete next PR sequence
113
114 1. **Docs PR:** land this tracker.
115 2. **Corpus portability PR:** rebase/portable manifest tooling plus missing-file smoke
test.
```

18. assistant (2026-07-03T16:40:05.616Z)

Now let me carefully check the dependency graph for any circular dependencies or invalid references. Let me create a systematic analysis:

19. user (2026-07-03T16:40:07.947Z)

```
A0: depends on - (start)
A1: depends on A0 (valid)
A2: depends on A0 (valid)
A3: depends on A2 (valid)
A4: depends on A2 (valid)
A5: depends on A2 (valid)
A6: depends on A2, A4 (valid - both depend on A2, no circle)
A7: depends on A2 (valid)
A8: depends on A6, A7 (valid - both A6 and A7 are downstream of A2)
A9: depends on A5, A8 (valid - A5 depends on A2, A8 depends on A6/A7)
A10: depends on A4, A5, A8 (valid - all are downstream of A2)
A11: depends on A10 (valid)
A12: depends on A1, A3 (A1 depends on A0, A3 depends on A2 - valid, both parallel)
A13: depends on A1, A4, A10, A11 (all valid - mixed lineages converge)
```

20. assistant (2026-07-03T16:40:09.790Z)

Now let me check the markdown links in the added sections:

21. user (2026-07-03T16:40:11.149Z)

(Bash completed with no output)

22. assistant (2026-07-03T16:40:13.278Z)

Let me check for markdown links more carefully:

23. user (2026-07-03T16:40:14.619Z)

```
7:> **Honesty note:** `[verified]` means checked against the local synced repos on 2026-07-03.
`[analysis]` means derived from a local command run on that snapshot. This is not legal,
financial, or launch approval.
13:The corpus is now large enough to move from "grow labels first" into alpha-readiness work,
but the 15-video headline is not itself a launch gate pass. `[verified]` The golden manifest has
15 badminton videos; `[analysis]` the local manifest needed path rebasing before manifest-driven
commands ran; `[analysis]` only 6 of 15 feature JSONs had the full fusion schema; `[analysis]`
the n=15 quality baseline is stale and must be intentionally refreshed before it can gate
anything.
```

24. assistant (2026-07-03T16:40:16.608Z)

Now let me verify the specific cross-references to other docs mentioned in the Relation and References sections:

25. user (2026-07-03T16:40:18.308Z)

```
-rw-r--r-- 1 avidu 197609 40690 Jun 25 16:24 docs/COMPUTE_DECOUPLED_SERVING/README.md
-rw-r--r-- 1 avidu 197609 4806 Jun 25 16:24 docs/data_pipeline/GOLDEN_VIDEOS.md
-rw-r--r-- 1 avidu 197609 23965 Jul 3 22:02 docs/NEXT_STEPS.md
-rw-r--r-- 1 avidu 197609 15067 Jun 25 16:24 docs/R2_EVAL_METRIC_DESIGN.md
-rw-r--r-- 1 avidu 197609 10265 Jul 2 08:13 docs/RALLY_DETECTION_FIXES_PLAN.md
-rw-r--r-- 1 avidu 197609 8260 Jun 25 16:24 docs/ROADMAP.md
```

26. assistant (2026-07-03T16:40:20.290Z)

Good, the referenced docs exist. Now let me look for specific internal contradictions or issues. Let me check the status fields carefully:

27. user (2026-07-03T16:40:20.770Z)

```
1      # Alpha Launch Readiness Plan
2
3      > **Status:** `DRAFT` - cross-repo execution tracker; **Owner:** @avidullu; **Created:**
2026-07-03; **Last updated:** 2026-07-03
4      > **Lifecycle:** `DRAFT` -> IN PROGRESS -> DONE -> archived`
5      > **Tracking anchors:** Section 7 is the progress tracker and source of truth; indexed
in `docs/README.md` and `docs/DOC_STATUS.md`.
6      > **Relation to existing docs:** consolidates the launch-relevant pending items from
`NEXT_STEPS.md`, `ROADMAP.md`, `data_pipeline/GOLDEN_VIDEOS.md`, `R2_EVAL_METRIC_DESIGN.md`,
`COMPUTE_DECOUPLED_SERVING/`, and the sibling `khelsutra` Studio docs.
7      > **Honesty note:** `[verified]` means checked against the local synced repos on
2026-07-03. `[analysis]` means derived from a local command run on that snapshot. This is not
legal, financial, or launch approval.
8
9      ---
10
```

28. assistant (2026-07-03T16:40:22.832Z)

Now let me carefully review the decisions section for any contradictions with later sections:

29. user (2026-07-03T16:40:22.961Z)

```
27      ## 2. Decisions locked
28
29      | # | Decision | Source / date | Implication |
30      |---|---|---|---|
31      | D1 | **Product alpha and owned-model launch are separate decisions.** |
`NEXT_STEPS.md` NO-GO banner, 2026-06-30; verified 2026-07-03 | We can launch a controlled
hosted product loop without claiming the owned model is commercially cleared. |
32      | D2 | **First alpha must be page-driven, not SSH/CLI-driven.** | `NEXT_STEPS.md` P0 UI-
driven/shareable item; verified 2026-07-03 | Hosted service, edge auth, upload, job polling,
compile, and download are launch-critical. |
33      | D3 | **The 15-video corpus is now a measurement asset, not a free launch pass.** |
`data_pipeline/GOLDEN_VIDEOS.md`; local eval runs, 2026-07-03 | Rebase/portability, feature
completeness, baseline refresh, and gate ratification come before quality claims. |
34      | D4 | **Do not promote served Gate A by default from the current n=15 evidence.** |
`backend.eval.serve_contrast` run, 2026-07-03 | Proposal-side lift does not survive served-
windowing safety; keep it as an opt-in/analysis lever. |
35      | D5 | **P10/P11 corpus promotion and train/re-eval are alpha-plus unless owner reopens
D13.** | sibling `khelsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`; verified 2026-07-03 | First alpha
can use manual/offline corpus promotion; automated flywheel follows after serving and gates are
stable. |
36
37      ## 3. Verified snapshot, 2026-07-03
38
39      All five Git checkouts were fast-forwarded before analysis:
40
```

```

41 | Repo | Branch | Synced HEAD |
42 |---|---|---|
43 | `badminton-highlight-indexer` | `master` | `e4914d1` |
44 | `khelsutra` | `master` | `b0f70a5` |
45 | `rally-annotator` | `main` | `ef9e088` |
46 | `sports-data-collector` | `master` | `bfc014f` |

```

30. assistant (2026-07-03T16:40:24.754Z)

Now let me check the Table in DOC_STATUS.md to verify consistency:

31. user (2026-07-03T16:40:25.110Z)

```

30    ### 3a. Tracked projects ? lifecycle docs (archive when DONE; each doc's $7 is the
completeness source-of-truth)
31    | Doc | Status | Archive-ready? | Next action |
32    |---|---|---|---|
33    | `ALPHA_LAUNCH_READINESS.md` | **DRAFT** ? cross-repo alpha readiness tracker created
2026-07-03 after the golden corpus reached n=15; separates hosted product-alpha work from owned-
model/commercial rally-detection gates | No | Execute $7 in order: corpus portability + n=15
baselines, hosted alpha serving, ablation/fusion reruns, then launch GO/NO-GO |
34    | `COMPUTE_DECOUPLED_SERVING/` | **IN PROGRESS** ? M0 approved + **M1?M9 LARGELY
SHIPPED** (#279?#290; M5 #286, M6 #287, M8 #288, M9-auth #290 all merged, default-OFF); Gen-0
weights pushed (`weights/0.1.0-l4/`) | No | M11 flywheel + M12 gdrive-api (owner residual:
counsel/prices/0Auth/real-run within the D17 cap) |

```

32. assistant (2026-07-03T16:40:27.694Z)

Good. Now let me verify there are no section numbering inconsistencies and check if there are any claims marked `[verified]` in section 0 that might be contradicted elsewhere. Let me look at the actual content more carefully:

33. user (2026-07-03T16:40:28.078Z)

```

49    Corpus/eval facts:
50
51    - `[verified]` `docs/data_pipeline/GOLDEN_VIDEOS.md` says
`output/human_lovo_manifest.json` is the ingested golden corpus at `n = 15`.
52    - `[analysis]` The local manifest had stale absolute paths. A temporary rebased manifest
found all 15 trajectories and labels in this workspace.
53    - `[analysis]` There are 15 `output/*_golden_features.json` files, but only 6 carry full
shuttle/person/audio candidate features. The newer 9 are windowing-only schema and are skipped
by fusion/person eval.
54    - `[analysis]` `python -m backend.eval.fusion_compare --features-dir output` loaded only
6 videos and measured person-fusion `Delta F1 = -0.021`; not promotable.
55    - `[analysis]` `python -m backend.eval.serve_contrast --manifest <rebased-n15>` measured
proposal Gate A mean `Delta F1 +0.0287`, but served mean `Delta F1 -0.0328`; not served-safe.
56    - `[analysis]` `python -m training.gen0.harness --manifest <rebased-n15> --floor
eval_baselines/heuristic_lovo_n6.json` produced learned LOVO mean F1 `0.551`, `ship=False`;
blockers remain stale n=6 floor/sign test, uncalibrated ship target, and WASB C3 provenance.
57    - `[analysis]` `python scripts/nightly_regression.py --manifest <rebased-n15>
--features-dir output --no-report` reported default/milestone tolF1 `0.3636` vs baseline-off
tolF1 `0.1382`, but marked the frozen baseline stale due to config/corpus changes.
58    - `[analysis]` `python -m backend.eval.ablation --features-dir output --granularity
group` currently fails at import due to a circular import between `backend/eval/ablation.py` and
`backend/eval/ablation_pdf.py`.
59
60    ## 4. Launch strategy
61
62    ### 4.1 Alpha lane: make the product usable
63
64    The alpha lane is successful when an invited tester can open the app, authenticate,
upload a video, watch the job progress, pick rallies, compile a reel, and download it. This lane
should avoid model-quality promises beyond the measured state and should default to the most

```


reliable serving path.

65

66 ### 4.2 Quality lane: make the 15-video corpus gate-ready

67

68 The quality lane is successful when the 15-video corpus is portable, reproducible, feature-complete, and backed by refreshed baselines. Only then should R2 tolF1, Gen-0/TemporalMaxer, or other model/promotion decisions be ratified.

34. assistant (2026-07-03T16:40:29.847Z)

Now let me carefully check the README.md link that was added:

35. user (2026-07-03T16:40:30.073Z)

17 ## Product / platform plans

18 - ALPHA_LAUNCH_READINESS.md ? *** Active tracked project ('DRAFT', created 2026-07-03):** cross-repo alpha readiness tracker after the golden corpus reached 15 videos. Separates the hosted product-alpha lane (shareable upload?process?pick?reel) from the owned-model/commercial rally-detection gate (ship target + WASB-C3 still unresolved). §7 is the source of truth for the concrete next PR sequence: corpus portability, n=15 baseline refresh, ablation/fusion reruns, hosted serving, alpha copy/launch review.

19 - [COMPUTE_DECOUPLED_SERVING/](COMPUTE_DECOUPLED_SERVING/README.md) ? *** Active tracked project (2026-06-21, 'DRAFT' owner-gated):** the productionization successor to DECOUPLED_COMPUTE (its deferred M3/M4). **One pure core (DETECT?WINDOW?STITCH) that never branches on deployment profile**, with a `ComputeBackend` (where it runs) + `StorageBackend` (what bytes) selected by **one startup config**. Ships **truly-local** (local/USB + local GPU, in-process stitch, zero cloud) + **cloud-serving** (upload<2GB / gdrive / bucket ? a **Cloud Run GPU job runs detect AND stitch**, metered into a per-job cost ledger) + **cpu-mock** (CI). Includes [TESTING_STRATEGY.md](COMPUTE_DECOUPLED_SERVING/TESTING_STRATEGY.md) (profile-PARITY proof for \$0), [architecture.svg](COMPUTE_DECOUPLED_SERVING/architecture.svg), a [runlogs/](COMPUTE_DECOUPLED_SERVING/runlogs/) posterity dir, and **ADR-014** ([lineage ADR-009?010?011?012?013?014](archives/decisions/DECISIONS.md)). §7 M0?M10 tracker is the source of truth.

36. assistant (2026-07-03T16:40:32.581Z)

Good. Now let me verify the link at line 18 of README.md actually resolves to the file:

37. user (2026-07-03T16:40:33.591Z)

-rw-r--r-- 1 avidu 197609 14883 Jul 3 22:03 ALPHA_LAUNCH_READINESS.md

Link resolves: OK

38. assistant (2026-07-03T16:40:35.912Z)

Now let me look carefully at the status/gated columns to check for inconsistencies. Let me examine each row in section 7:

39. user (2026-07-03T16:40:37.234Z)

1: ID= ID Status= Gated? Gated= Depends on
2: ID=---- Status=----- Gated=-----
3: ID= A0 Status= No Gated= -
4: ID= A1 Status= Owner spend / CF config Gated= A0
5: ID= A2 Status= No Gated= A0
6: ID= A3 Status= Owner upload/storage scope Gated= A2
7: ID= A4 Status= Reviewer sign-off Gated= A2
8: ID= A5 Status= No Gated= A2
9: ID= A6 Status= No Gated= A2, A4
10: ID= A7 Status= GPU/data availability Gated= A2
11: ID= A8 Status= Reviewer sign-off Gated= A6, A7
12: ID= A9 Status= Owner approval for M0.4 scope Gated= A5, A8
13: ID= A10 Status= Owner decision Gated= A4, A5, A8
14: ID= A11 Status= Legal/provenance Gated= A10

15: ID= A12 Status= A1, A3 Gated= `khelsutra` + `badminton-highlight-indexer` + collector
16: ID= A13 Status= Owner launch decision Gated= A1, A4, A10, A11

40. assistant (2026-07-03T16:40:39.781Z)

I see the issue - the awk parsing is misaligned. Let me view the table more directly:

41. user (2026-07-03T16:40:40.185Z)

```
95      | ID | Deliverable | Repo / owner | Depends on | Gated? | Status | PR |
96      |----|-----|-----|-----|-----|-----|----|
97      | A0 | Create this alpha readiness tracker and index it in docs | `badminton-highlight-
indexer` | - | No | IN PROGRESS | #451 |
98      | A1 | Stand up alpha serving endpoint: Cloud Run/GPU or approved host, health check,
edge auth ON, upload -> process -> jobs -> compile -> download verified | `khelsutra` +
`badminton-highlight-indexer` | A0 | Owner spend / CF config | TODO | - |
99      | A2 | Make the n=15 golden manifest portable/reproducible; add a path/corpus smoke
check that fails loudly when labels or trajectories are missing | `badminton-highlight-indexer`
| A0 | No | TODO | - |
100     | A3 | Promote the 15-video source-video/MD5 records into the collector/vault path;
reconcile collector's six-video source manifest with the 15-video eval corpus | `sports-data-
collector` + vault | A2 | Owner upload/storage scope | TODO | - |
101     | A4 | Refresh the n=15 nightly regression baseline intentionally and record the exact
command/output; keep stale-baseline warnings until reviewed | `badminton-highlight-indexer` | A2
| Reviewer sign-off | TODO | - |
102     | A5 | Recompute the heuristic n=15 floor and replace the stale `heuristic_lovo_n6`
floor for future Gen-0 comparisons | `badminton-highlight-indexer` | A2 | No | TODO | - |
103     | A6 | Fix `backend.eval.ablation` CLI circular import, then run the R2/tolF1 ablation
on the n=15 corpus | `badminton-highlight-indexer` | A2, A4 | No | TODO | - |
104     | A7 | Regenerate or upconvert full-fusion feature JSONs for the newer 9 videos so
fusion/person/audio analyses run over all 15 | `badminton-highlight-indexer` | A2 | GPU/data
availability | TODO | - |
105     | A8 | Re-run `fusion_compare`, group ablation, and served contrast after A7; explicitly
promote/reject person fusion, Gate A served-default, and any other R-series default flips |
`badminton-highlight-indexer` | A6, A7 | Reviewer sign-off | TODO | - |
106     | A9 | Run Gen-0/TemporalMaxer shadow training against the refreshed n=15 floor; produce
a non-serving artifact bundle and gate verdict | `badminton-highlight-indexer` | A5, A8 | Owner
approval for M0.4 scope | TODO | - |
107     | A10 | Decide and record the alpha ship-gate target: metric, corpus slice, minimum
threshold, significance rule, and what claims are allowed | Owner + `badminton-highlight-
indexer` | A4, A5, A8 | Owner decision | GATED | - |
108     | A11 | Resolve or explicitly defer WASB C3 provenance for alpha messaging; if deferred,
document "no commercial model sharing" copy guardrail | Owner/legal | A10 | Legal/provenance |
GATED | - |
109     | A12 | Reopen or defer P10/P11 corpus flywheel: `PATCH /api/segments`, collector
promotion, `train|reeval` job, and delta-F1 surface | `khelsutra` + `badminton-highlight-
indexer` + collector | A1, A3 | Owner D13 reopen | GATED | - |
110     | A13 | Alpha launch review: runbook, privacy/consent copy, known limits,
rollback/disable steps, and owner GO/NO-GO | Cross-repo | A1, A4, A10, A11 | Owner launch
decision | TODO | - |
```

42. assistant (2026-07-03T16:40:42.995Z)

Now let me check for potential content contradictions by looking at claims in section 0 (TL;DR)
against the facts in section 3:

43. user (2026-07-03T16:40:43.620Z)

```
11      ## 0. TL;DR
12
13      The corpus is now large enough to move from "grow labels first" into alpha-readiness
work, but the 15-video headline is not itself a launch gate pass. `[verified]` The golden
manifest has 15 badminton videos; `[analysis]` the local manifest needed path rebasing before
manifest-driven commands ran; `[analysis]` only 6 of 15 feature JSONs had the full fusion
```

schema; `[analysis]` the n=15 quality baseline is stale and must be intentionally refreshed before it can gate anything.

14

15 The first controlled alpha should be a hosted, authenticated product loop: upload a video, process it, select rallies, compile, and download a reel. The owned-model/commercial rally-detection claim remains a separate NO-GO until the ship-gate target is ratified and the WASB checkpoint provenance is resolved.

16

44. assistant (2026-07-03T16:40:45.879Z)

Now let me check the section 2 (Decisions locked) reference to D13 in section 5:

45. user (2026-07-03T16:40:46.170Z)

```
1      # Alpha Launch Readiness Plan
2
3      > **Status:** `DRAFT` - cross-repo execution tracker; **Owner:** @avidullu; **Created:**
2026-07-03; **Last updated:** 2026-07-03
4      > **Lifecycle:** `DRAFT` -> IN PROGRESS -> DONE -> archived`
5      > **Tracking anchors:** Section 7 is the progress tracker and source of truth; indexed
in `docs/README.md` and `docs/DOC_STATUS.md`.
6      > **Relation to existing docs:** consolidates the launch-relevant pending items from
`NEXT_STEPS.md`, `ROADMAP.md`, `data_pipeline/GOLDEN_VIDEOS.md`, `R2_EVAL_METRIC_DESIGN.md`,
`COMPUTE_DECOUPLED_SERVING/`, and the sibling `khealsutra` Studio docs.
7      > **Honesty note:** `[verified]` means checked against the local synced repos on
2026-07-03. `[analysis]` means derived from a local command run on that snapshot. This is not
legal, financial, or launch approval.
8
9      ---
10
```

46. assistant (2026-07-03T16:40:48.289Z)

Let me check if D13 is actually defined in Section 2:

47. user (2026-07-03T16:40:49.162Z)

```
35:| D5 | **P10/P11 corpus promotion and train/re-eval are alpha-plus unless owner reopens
D13.** | sibling `khealsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`; verified 2026-07-03 | First alpha
can use manual/offline corpus promotion; automated flywheel follows after serving and gates are
stable. |
109:| A12 | Reopen or defer P10/P11 corpus flywheel: `PATCH /api/segments`, collector promotion,
`train|reeval` job, and delta-F1 surface | `khealsutra` + `badminton-highlight-indexer` +
collector | A1, A3 | Owner D13 reopen | GATED | - |
```

48. assistant (2026-07-03T16:40:52.410Z)

Found an issue! D13 is referenced in lines 35 and 109 but is never defined in Section 2 (Decisions locked). Section 2 only has D1-D5. Let me verify this more carefully:

49. user (2026-07-03T16:40:53.360Z)

2. Decisions locked

```
| # | Decision | Source / date | Implication |
|---|---|---|---|
| D1 | **Product alpha and owned-model launch are separate decisions.** | `NEXT_STEPS.md` NO-GO
banner, 2026-06-30; verified 2026-07-03 | We can launch a controlled hosted product loop without
claiming the owned model is commercially cleared. |
| D2 | **First alpha must be page-driven, not SSH/CLI-driven.** | `NEXT_STEPS.md` P0 UI-
driven/shareable item; verified 2026-07-03 | Hosted service, edge auth, upload, job polling,
compile, and download are launch-critical. |
```

```
| D3 | **The 15-video corpus is now a measurement asset, not a free launch pass.** |  
`data_pipeline/GOLDEN_VIDEOS.md`; local eval runs, 2026-07-03 | Rebase/portability, feature  
completeness, baseline refresh, and gate ratification come before quality claims. |  
| D4 | **Do not promote served Gate A by default from the current n=15 evidence.** |  
`backend.eval.serve_contrast` run, 2026-07-03 | Proposal-side lift does not survive served-  
windowing safety; keep it as an opt-in/analysis lever. |  
| D5 | **P10/P11 corpus promotion and train/re-eval are alpha-plus unless owner reopens D13.** |  
sibling `khelesutra/docs/STUDIO_MEDIA_LIFECYCLE.md`; verified 2026-07-03 | First alpha can use  
manual/offline corpus promotion; automated flywheel follows after serving and gates are stable.  
|
```

50. assistant (2026-07-03T16:40:56.470Z)

This is a clear issue. D13 is referenced in D5's "Implication" column and in A12's "Gated?" column, but D13 is not defined anywhere in Section 2. This is a broken cross-reference.

Let me check if this might be a reference to a different doc by searching NEXT_STEPS.md:

51. user (2026-07-03T16:40:57.420Z)

```
12:0. **? [owner-priority, 2026-06-21] MAKE IT UI-DRIVEN + ALPHA-SHAREABLE ? "a few recordings  
lying around ? highlights, from a page, no SSH".** Today the cloud flow is **SSH/CLI-only**  
(`docs/CLOUD_GPU_DRYRUN_GUIDE.md`); **M12 does NOT fix it.** Needs (a separate khelesutra `web`/  
SPA + deploy track, NOT an engine seam): **(i)** deploy the engine as a reachable **service**  
(M7 ? Cloud Run service, ? owner GCP spend for a *persistent* endpoint), **(ii)** the **SPA  
ingest/progress screen** (khelesutra `web` B-P2: enter/upload video ? `/api/process` ? poll  
`/api/jobs` ? reel; ZERO engine code), **(iii)** flip **M9 edge-auth ON** (merged #290, default-  
OFF) to gate alpha tenants, **(iv)** a **compute PICKER in the UI** ("your machine vs our  
cloud", D13/D15). The engine seams make it *possible*; the SPA makes it *usable*. Full record:  
[[ui-driven-cloud-serving-gap]]. **Concrete execution tracker after the n=15 corpus review:**  
`docs/ALPHA_LAUNCH_READINESS.md` §7.
```

52. assistant (2026-07-03T16:41:00.516Z)

So D13 is mentioned in NEXT_STEPS.md but not defined. This is an external cross-repo reference issue. Let me also check whether "D13" should reference something that exists in this doc or elsewhere. Let me check if there's a mistake in line 35 by looking at what D5 says:

D5 states: "P10/P11 corpus promotion and train/re-eval are alpha-plus unless owner reopens D13." The mention of D13 here is undefined in this document. The same undefined D13 appears in A12's "Gated?" column.

Let me verify whether this is the only issue by scanning all the facts marked `[verified]`:

53. user (2026-07-03T16:41:01.358Z)

```
7:> **Honesty note:** `[verified]` means checked against the local synced repos on 2026-07-03.  
`[analysis]` means derived from a local command run on that snapshot. This is not legal,  
financial, or launch approval.  
13:The corpus is now large enough to move from "grow labels first" into alpha-readiness work,  
but the 15-video headline is not itself a launch gate pass. `[verified]` The golden manifest has  
15 badminton videos; `[analysis]` the local manifest needed path rebasing before manifest-driven  
commands ran; `[analysis]` only 6 of 15 feature JSONs had the full fusion schema; `[analysis]`  
the n=15 quality baseline is stale and must be intentionally refreshed before it can gate  
anything.  
51:- `[verified]` `docs/data_pipeline/GOLDEN_VIDEOS.md` says `output/human_lovo_manifest.json`  
is the ingested golden corpus at `n = 15`.
```

54. assistant (2026-07-03T16:41:04.616Z)

Now let me verify that claim at line 51:

55. user (2026-07-03T16:41:05.558Z)

(`gts` + candidates), append a row to `output/human\_lovo\_manifest.json`. (`scratch/at\_par.py` scores a

Ingested golden corpus ? `output/human\_lovo\_manifest.json` (n = 15)

| GX030094 | badminton | 41 | **29.97** | `~/Golden Label ? /[Done] Video 6 - Badminton/GX030094\_proxy.mp4` | **ingested** (in `human\_lovo\_manifest.json`) |
| Badminton BXH 2 | badminton (multicourt) | 44 | 59.94 | `~/[Done] Video 10 - Badminton Multicourt/Badminton BXH 2\_proxy.mp4` | **ingested** (in `human\_lovo\_manifest.json`) |

56. assistant (2026-07-03T16:41:08.026Z)

Good, that claim verifies. Now let me check for any internal contradictions in the document. Let me look at section 5 Risks:

57. user (2026-07-03T16:41:08.769Z)

```
74      ## 5. Risks and guardrails
75
76      | Risk | Why it matters | Guardrail |
77      |---|---|---|
78      | Launch activity implies model launch approval | The engine docs explicitly say rally-
detection product launch is NO-GO until ship target and WASB C3 clear | Keep all alpha copy
scoped to invite-only processing, not owned-model claims |
79      | Stale local manifests create false confidence | Manifest-driven tools skipped or
failed until paths were rebased | Make the corpus manifest portable and add a path/fixture smoke
check |
80      | 15-video corpus is only partially feature-complete | Fusion/person/audio eval is still
n=6 today | Regenerate or upconvert full-fusion features for all 15 before fusion decisions |
81      | Baseline refresh accidentally rubber-stamps a regression | Nightly runner correctly
reported stale baselines | Refresh baselines only after reviewer sign-off and record the
command/output |
82      | P10/P11 expands scope before hosted alpha is usable | Corpus flywheel is valuable but
can consume the alpha launch window | Keep P10/P11 behind explicit owner reopen unless alpha
lane is already stable |
83      | WASB checkpoint provenance remains unresolved | Blocks commercial sharing of a trained
model regardless of F1 | Treat C3 as a hard model-release gate; do not hide it inside product
launch tasks |
84
```

58. assistant (2026-07-03T16:41:11.256Z)

Let me also double-check the README.md link target is correct and exists at the right location:

59. user (2026-07-03T16:41:12.494Z)

ALPHA_LAUNCH_READINESS.md: ASCII text, with very long lines (479)

Alpha Launch Readiness Plan

60. assistant (2026-07-03T16:41:14.738Z)

Good. Now let me check for any potential markdown table format issues. Let me verify there are no broken markdown tables:

61. user (2026-07-03T16:41:20.812Z)

Line 80: 4 pipes - | 15-video corpus is only partially feature-complete | Fusion/person/audio eval
Line 81: 4 pipes - | Baseline refresh accidentally rubber-stamps a regression | Nightly runner corr
Line 82: 4 pipes - | P10/P11 expands scope before hosted alpha is usable | Corpus flywheel is valua
Line 83: 4 pipes - | WASB checkpoint provenance remains unresolved | Blocks commercial sharing of a
Line 95: 8 pipes - | ID | Deliverable | Repo / owner | Depends on | Gated? | Status | PR |

```

Line 96: 8 pipes - |----|-----|-----|-----|-----|-----|----|
Line 97: 8 pipes - | A0 | Create this alpha readiness tracker and index it in docs | `badminton-
high
Line 98: 8 pipes - | A1 | Stand up alpha serving endpoint: Cloud Run/GPU or approved host,
health c
Line 99: 8 pipes - | A2 | Make the n=15 golden manifest portable/reproducible; add a path/corpus
sm
Line 100: 8 pipes - | A3 | Promote the 15-video source-video/MD5 records into the
collector/vault pa
Line 101: 8 pipes - | A4 | Refresh the n=15 nightly regression baseline intentionally and record
the
Line 102: 8 pipes - | A5 | Recompute the heuristic n=15 floor and replace the stale
`heuristic_lovo_
Line 103: 8 pipes - | A6 | Fix `backend.eval.ablation` CLI circular import, then run the
R2/tolF1 ab
Line 104: 8 pipes - | A7 | Regenerate or upconvert full-fusion feature JSONs for the newer 9
videos
Line 105: 8 pipes - | A8 | Re-run `fusion_compare`, group ablation, and served contrast after
A7; ex
Line 106: 8 pipes - | A9 | Run Gen-0/TemporalMaxer shadow training against the refreshed n=15
floor;
Line 107: 8 pipes - | A10 | Decide and record the alpha ship-gate target: metric, corpus slice,
mini
Line 108: 8 pipes - | A11 | Resolve or explicitly defer WASB C3 provenance for alpha messaging;
if d
Line 109: 9 pipes - | A12 | Reopen or defer P10/P11 corpus flywheel: `PATCH /api/segments`,
collecto
Line 110: 8 pipes - | A13 | Alpha launch review: runbook, privacy/consent copy, known limits,
rollba

```

62. assistant (2026-07-03T16:41:23.349Z)

I found another issue! Line 109 (A12 row) has 9 pipes while all other rows in the same table have 8 pipes. This is a markdown table format error. Let me verify this:

63. user (2026-07-03T16:41:23.799Z)

```

108      | A11 | Resolve or explicitly defer WASB C3 provenance for alpha messaging; if deferred,
document "no commercial model sharing" copy guardrail | Owner/legal | A10 | Legal/provenance |
GATED | - |
109      | A12 | Reopen or defer P10/P11 corpus flywheel: `PATCH /api/segments`, collector
promotion, `train|reeval` job, and delta-F1 surface | `khelsutra` + `badminton-highlight-
indexer` + collector | A1, A3 | Owner D13 reopen | GATED | - |
110      | A13 | Alpha launch review: runbook, privacy/consent copy, known limits,
rollback/disable steps, and owner GO/NO-GO | Cross-repo | A1, A4, A10, A11 | Owner launch
decision | TODO | - |

```

64. assistant (2026-07-03T16:41:26.890Z)

Let me count the pipes carefully in line 109:

```

...
| A12 | Reopen or defer P10/P11 corpus flywheel: `PATCH /api/segments`, collector promotion,
`train|reeval` job, and delta-F1 surface | `khelsutra` + `badminton-highlight-indexer` +
collector | A1, A3 | Owner D13 reopen | GATED | - |
...

```

The issue is the backtick in `train|reeval` - the pipe character inside the backticks is being counted as a pipe! But actually, this should be fine in markdown tables - the pipes inside backticks shouldn't break the table. Let me verify this is actually a rendering issue or not:

65. user (2026-07-03T16:41:29.100Z)

Line 109:

```
'| A12 | Reopen or defer P10/P11 corpus flywheel: `PATCH /api/segments`, collector promotion,
`train|reeval` job, and delta-F1 surface | `khelsutra` + `badminton-highlight-indexer` +
collector | A1, A3 | Owner D13 reopen | GATED | - |\n'
```

Pipe count: 9

Backtick sections:

```
`PATCH /api/segments`
`train|reeval`
  -> Contains pipe
`khelsutra`
`badminton-highlight-indexer`
```

66. assistant (2026-07-03T16:41:33.959Z)

So the `train|reeval` code snippet contains a literal pipe character within backticks. In standard markdown table syntax, this can cause rendering issues because the pipe inside backticks might still be interpreted as a column separator. This is a broken markdown table issue.

Let me now compile all my findings. I've identified:

1. ****Line 35 (D5) and Line 109 (A12)**:** Reference to undefined "D13" - D13 is not defined anywhere in Section 2 (Decisions locked), which only defines D1-D5. This is a broken cross-reference.
2. ****Line 109 (A12 row in table)**:** The pipe character inside the backtick `` `train|reeval` `` will break the markdown table rendering, as markdown table parsers treat pipes inside backticks as column separators.

Let me verify there are no other similar issues:

67. user (2026-07-03T16:41:36.063Z)

Line 109: Pipe in backticks: `train|reeval`

68. assistant (2026-07-03T16:41:41.358Z)

Perfect. Now I have my complete findings. Let me create the final JSON array report:

```
```json
[
 {
 "file": "docs/ALPHA_LAUNCH_READINESS.md",
 "line": 35,
 "summary": "Undefined cross-reference D13 in decision D5",
 "failure_scenario": "D5 states 'P10/P11 corpus promotion and train/re-eval are alpha-plus unless owner reopens D13.' But D13 is never defined in Section 2 (Decisions locked), which only includes D1-D5. A reader following the dependency chain will encounter this undefined reference and won't know what decision D13 refers to."
 },
 {
 "file": "docs/ALPHA_LAUNCH_READINESS.md",
 "line": 109,
 "summary": "Pipe character inside backticks breaks markdown table rendering",
 "failure_scenario": "The A12 row contains backtick code snippet `train|reeval` with a literal pipe character. Markdown table parsers treat pipes as column separators even inside backticks, causing the table to render with 9 columns instead of 8. This will misalign the table and display incorrectly."
 },
 {
 "file": "docs/ALPHA_LAUNCH_READINESS.md",
 "line": 109,
 "summary": "Undefined cross-reference D13 in A12 gating condition",

```

```
 "failure_scenario": "A12's 'Gated?' column states 'Owner D13 reopen', referencing the
undefined decision D13. This creates the same issue as D5: readers cannot determine what
condition gates A12 without knowing what D13 is."
 }
]
...
```